

Tight Sample Complexity Bounds for Entropic Best Policy Identification

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Editor: Under Review for COLT 2026

Abstract

This is a great paper and it has a concise abstract.

Keywords: List of keywords

1. Introduction

2. Risk-Sensitive Episodic Reinforcement Learning

We consider a finite-horizon MDP $M = (\mathcal{S}, \mathcal{A}, H, \{r_h\}_{h=1}^H, \{p_h\}_{h=1}^H)$ where $\mathcal{S}$ and $\mathcal{A}$ denote the finite state and action spaces, respectively, $H \in \mathbb{N}$ is the fixed horizon length, $p_h(\cdot \mid s, a)$ is the transition kernel at step h for each $h \in \{1, \dots, H\}$, and $r_h : \mathcal{S} \times \mathcal{A} \rightarrow [0, 1]$ is the (deterministic) reward function at step h , which we assume to be bounded in $[0, 1]$.

In the learning problem, the transition kernels $\{p_h\}_{h=1}^H$ are unknown to the learner, who must interact with the environment. Throughout this interaction, the learner must devise a policy π , that is, a (possibly non-stationary) mapping prescribing which action to take in each state (and step), with the goal of maximizing a suitable performance criterion. In the literature, three standard interaction models with the environment are generally distinguished.

- **Generative model** (Mortensen and Talebi, 2025). In the generative model (also called a sampling oracle), the learner has access to an oracle such that, for any query $(s, a, h) \in \mathcal{S} \times \mathcal{A} \times \{1, \dots, H\}$, the oracle returns a sample (r, s') where $r = r_h(s, a)$ and $s' \sim p_h(\cdot \mid s, a)$. Each call to the oracle is independent of the past, and the learner may choose its queries adaptively based on all previously observed samples.
- **Online (forward) model** In the forward interaction model, the learner interacts with the MDP over episodes. At the beginning of each episode t , an initial state s_1^t is drawn from a fixed initial distribution μ_1 on $\mathcal{S}$. At each step $h \in \{1, \dots, H\}$ of episode t , the learner observes the current state s_h^t , selects an action $a_h^k \in \mathcal{A}$ according to some policy π_t , then receives a reward $r_h(s_h^k, a_h^k)$ and the environment goes to the next state $s_{h+1}^k \sim p_h(\cdot \mid s_h^k, a_h^k)$
- **Reward-free model** In the reward-free model, the learner interacts with the environment as in the forward (online) model but does not observe any rewards during the exploration phase. The goal is to design an exploration strategy that visits all relevant state–action pairs sufficiently often so that, once a reward function $r = \{r_h\}_{h=1}^H$ is revealed, the learner must then output a policy based on the data collected during exploration

In this work, we focus exclusively on the online (forward) interaction model. Moreover, rather than the standard risk-neutral objective where we optimize the total expected reward along the trajectory, we adopt a risk-sensitive performance criterion based on the entropic risk measure which reflects the agent’s attitude toward uncertainty, allowing for both risk-seeking and risk-averse behavior.

Policy & value functions A deterministic policy π is a collection of functions $\pi_h : \mathcal{S} \rightarrow \mathcal{A}$ for all $h \in [H]$, where every π_h maps each state to a single action. The entropic value functions of π , denoted by V_h^π , are defined as

$$V_h^\pi(s) \triangleq \frac{1}{\beta} \log \mathbb{E} \left[\exp \left(\beta \sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'}) \right) \mid s_h = s \right]$$

where $a_{h'} \triangleq \pi_{h'}(s_{h'})$ and $s_{h'+1} \sim p_{h'}(\cdot \mid s_{h'}, a_{h'})$

The optimal entropic value functions are defined as $V_h^*(s) \triangleq \sup_{\pi} V_h^\pi(s)$ for $h \in [H]$. As the case with the total reward expectation, both V_h^π and V_h^* satisfy the Bellman equation (Borkar and Meyn, 2000) and can be expressed using the entropic Q-value functions Q_h^π and Q_h^* in the following way:

$$V_h^\pi(s) = Q_h^\pi(s, \pi_h(s)), \quad \text{with} \quad Q_h^\pi(s, a) \triangleq \frac{1}{\beta} \log \mathbb{E}_{S' \sim p_h(\cdot \mid s, a)} [\exp(\beta(r_h(s, a) + V_{h+1}^\pi(S')))] \quad \text{and,}$$

$$V_h^*(s) = \max_{a \in \mathcal{A}} Q_h^*(s, a), \quad \text{with} \quad Q_h^*(s, a) \triangleq \frac{1}{\beta} \log \mathbb{E}_{S' \sim p_h(\cdot \mid s, a)} [\exp(\beta(r_h(s, a) + V_{h+1}^*(S')))] ,$$

with terminal condition $V_{H+1}^\pi(s) = V_{H+1}^*(s) = 0$ for all $s \in \mathcal{S}$. For any bounded function $f : \mathcal{S} \rightarrow \mathbb{R}$, we denote by $(P_h f)(s, a) \triangleq \mathbb{E}_{S' \sim p_h(\cdot \mid s, a)} [f(S')]$ the action of the Markov kernel p_h on f . For any function $g : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$ and (possibly stochastic) policy π_h , we write $(\pi_h g)(s) \triangleq \mathbb{E}_{A \sim \pi_h(\cdot \mid s)} [g(s, A)]$ for the composition with the policy at step h .

Empirical MDP. Let $(s_h^i, a_h^i, s_{h+1}^i)$ denote the state, action, and subsequent state observed by the algorithm at step h of episode i . For any $h \in [H]$ and any $(s, a) \in \mathcal{S} \times \mathcal{A}$, define the visitation counts

$$n_h^t(s, a) \triangleq \sum_{i=1}^t \mathbf{1}\{(s_h^i, a_h^i) = (s, a)\} \quad n_h^t(s, a, s') \triangleq \sum_{i=1}^t \mathbf{1}\{(s_h^i, a_h^i, s_{h+1}^i) = (s, a, s')\}$$

These quantities induce the empirical transition probabilities

$$\tilde{p}_h^t(s' \mid s, a) \triangleq \begin{cases} \frac{n_h^t(s, a, s')}{n_h^t(s, a)} & \text{if } n_h^t(s, a) > 0, \\ \frac{1}{|\mathcal{S}|} & \text{otherwise} \end{cases}$$

We also denote $\bar{n}_h^t(s, a) = \mathbb{E}[n_h^t(s, a)]$ as the expected number of empirical visits, which is called the pseudo-counts.

Best Policy Identification (BPI) under entropic risk We are interested in this work in the Best policy identification problem. In each episode t , the agent follows a policy π^t (the sampling rule) based only on the information collected up to and including episode $t - 1$. At the end of each

episode, the agent can decide to stop collecting data (we denote by τ its random stopping time) and outputs a guess $\hat{\pi}$ for the optimal policy.

A BPI algorithm is therefore made of a triple $((\pi^t)_{t \in \mathbb{N}}, \tau, \hat{\pi})$. The goal is to build an (ε, δ) -PAC algorithm according to the following definition, for which the *sample complexity*, that is the number of exploration episodes τ , is as small as possible.

Definition 1 (PAC algorithm for BPI) *An algorithm is (ε, δ) -PAC for best policy identification if it returns a policy $\hat{\pi}$ after some number of episodes τ that satisfies*

$$\mathbb{P} \left(V_1^*(s_1) - V_1^{\hat{\pi}}(s_1) \leq \varepsilon \right) \geq 1 - \delta$$

2.1. Lower bounds and Open Problem

For the entropic risk criterion, the exponential transformation $e^{\beta R}$ amplifies tail trajectories—up-weighting rare, high-return paths when $\beta > 0$ (or high-cost paths in cost form) so if the optimal performance relies on a hard state–action–time pair (s^*, a^*, h^*) with tiny reachability, the learner must repeatedly realize that rare transition for the exponentially weighted estimates to stabilize, which inflates estimator variance and induces hardness that grows roughly like $e^{|\beta|H}$, making learning strictly harder than in the risk-neutral case.

Theorem 2 *[PAC lower bound for the entropic value objective] Fix $\beta \in \mathbb{R}$ and integers $H \geq 3$, $S \geq 4$, $A \geq 2$, $\delta \in (0, \frac{S-3}{2})$, and sufficiently small $\varepsilon > 0$. There exists an episodic MDP with horizon H , S states, and A actions such that any (ε, δ) -PAC algorithm must, for some instance of this MDP, use an expected number of episodes T satisfying*

$$T = \Omega \left(SA \frac{e^{|\beta|H} - 1}{\beta^2 \varepsilon^2} \log \left(\frac{S}{\delta} \right) \right).$$

Our lower-bound construction is conceptually related to the lower bound of Mortensen and Talebi [Mortensen and Talebi \(2025\)](#), but the settings are different. First, Mortensen and Talebi work in a discounted MDP with access to a generative model, whereas we consider a finite-horizon episodic MDP in the forward online interaction model, where the learner only observes trajectories generated by its actions from the initial distribution. In their setting, the performance criteria are global value- or policy-learning guarantees such as $\|Q^* - Q_U\|_\infty \leq \varepsilon$ or $\|V^* - V^{\pi_U}\|_\infty \leq \varepsilon$, and the hard MDP can be viewed as a collection of independent Bernoulli estimation problems, one for each state-action pair: each generative query at a given (s, a) corresponds to observing a Bernoulli outcome indicating, for example, a transition to a “good” or “bad” absorbing state. An under-sampled pair then directly prevents a uniform accuracy guarantee via a standard change-of-measure argument applied independently to each such Bernoulli problem ([Mortensen and Talebi, 2025](#), Sec. 4). In our finite-horizon entropic BPI setting with forward interaction, performance is instead assessed via the entropic value of the entire H -step return from the initial distribution. A suboptimal decision in one of the bandits affects the criterion only through its contribution to this aggregated entropic transform, so we relate the fraction of unsolved bandits to the entropic value gap at the start state. This averaging effect over bandits, together with the non-linearity of the entropic risk, leads to our lower bound.

Proof [sketch of proof] We introduce the following MDP:

We start at state s_0 . We pick a random number uniformly from $\{1, \dots, n\}$ where $n = S - 3$ and we

jump to bandit i . The learned take an action $a \in \{1, \dots, A\}$ to jump either to $+$ to receive reward 1 or jump to $-$ and receive reward 0 and stays there for the rest of the episode. That means that if we take an optimal choice we receive a reward $H - 2$ at the end, otherwise we receive 0. For each

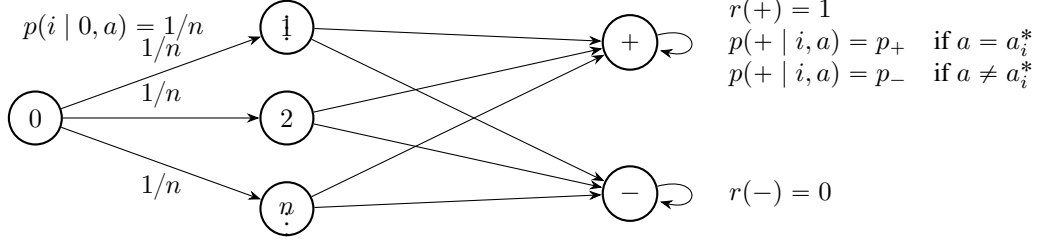


Figure 1: MDP with n parallel bandits: from state 0 we jump uniformly to bandit i , then an action leads to absorbing state $+$ (with reward 1) or $-$ (with reward 0), with success probabilities p_+ or p_- depending on whether a_i^* is played.

bandit i there is one optimal arm a_i^* that takes us to $+$ with probability p_+ and $A - 1$ arms that takes us to $-$ with probability p_- . Which arm is optimal for i define a vector $I \in [A]^n$. We constructed a family of MDPs and each MDP is identifiable by the vector I i.e which arms are optimal for each bandit.

The state-value function is : For the entropic risk-sensitive objective,

$$V(p) = \frac{1}{\beta} \log \mathbb{E}[e^{\beta R}] = \frac{1}{\beta} \log(1 + cp), \quad c := e^{\beta(H-2)} - 1,$$

where p denotes the success probability of the chosen arm.

Each bandit i has one optimal arm with success probability p_+ and $A - 1$ suboptimal arms with probability p_- . To make the instances hard to distinguish we chose

$$p_- = \frac{1}{2c} \quad p_+ = p_-(1 + \eta) \quad \eta = 3(e^{\beta\epsilon/H} - 1),$$

so that the entropic value gap satisfies $V(p_+) - V(p_-) = \epsilon$.

Let M^π denote the number of bandits on which a policy π selects a suboptimal arm, and define $\rho = M^\pi/n$. Then

$$V^* - V^\pi = -\frac{1}{\beta} \log(1 - (1 - e^{-\beta\epsilon/H})\rho).$$

If π is ϵ -optimal, then $V^* - V^\pi \leq \epsilon$, which implies

$$\rho \leq \frac{1 - e^{-\beta\epsilon}}{1 - e^{-\beta\epsilon/H}}, \quad \phi := 1 - \rho$$

for all sufficiently small ϵ . Hence, any ϵ -optimal policy must correctly solve at least $\phi n = \Theta(n)$ bandits. We then require that each such “must-solve” bandit is solved with error at most $\delta_i = \frac{\delta}{\phi n}$. A union bound ensures all these bandits are solved simultaneously with probability at least $1 - \delta$. Fix a must-solve bandit i . For each suboptimal arm b , consider an alternative instance where b is optimal. Applying a change-of-measure argument [Kaufmann et al. \(2016\)](#) yields

$$\mathbb{E}[N_{i,b}] \geq \frac{D_{\text{KL}}(1 - \delta_i, \delta_i)}{D_{\text{KL}}(\text{Ber}(p_-) \parallel \text{Ber}(p_+)})$$

Summing over the $A - 1$ suboptimal arms,

$$\mathbb{E}[N_i] \geq (A - 1) \frac{D_{\text{KL}}(1 - \delta_i, \delta_i)}{D_{\text{KL}}(\text{Ber}(p_-) \parallel \text{Ber}(p_+))}$$

Since each episode involves exactly one arm pull,

$$T = \sum_{i=1}^n \mathbb{E}[N_i] \geq \phi n (A - 1) \frac{D_{\text{KL}}(1 - \delta_i, \delta_i)}{D_{\text{KL}}(\text{Ber}(p_-) \parallel \text{Ber}(p_+))}$$

Using $D_{\text{KL}}(1 - x, x) \geq \log(1/(2x))$ for $x \leq 1/2$, and $D_{\text{KL}}(\text{Ber}(p_-) \parallel \text{Ber}(p_+)) \leq \frac{\eta^2}{2c-1}$ we obtain

$$T \geq \phi(S - 3)(A - 1) \frac{2(e^{\beta(H-2)} - 1) - 1}{\eta^2} \log \left(\frac{\phi(S - 3)}{2\delta} \right).$$

Finally, since $\eta = 3(e^{\beta\varepsilon} - 1) = \Theta(\beta\varepsilon)$ for ε small enough, we conclude after calculation

$$T = \Omega \left((S - 3)(A - 1) \frac{e^{\beta(H-2)}}{\beta^2 \varepsilon^2} \log \frac{S}{\delta} \right)$$

■

3. Algorithm and Matching Upper Bound

We now complement the lower bound (Theorem 2) with an algorithm whose sample complexity matches it up to logarithmic factors. A central difficulty with the entropic risk measure is that, unlike the risk-neutral objective, it is neither additive nor sub-additive, which complicates both algorithm design and analysis. In particular, standard UCB-style approaches rely on concentration for additive returns, whereas the entropic criterion involves a log-moment generating function and does not directly fit into risk-neutral concentration framework.

A common workaround is to use the Lipschitzness of the logarithm to reduce the analysis to risk-neutral quantities; however, this can yield coarse bounds and, more importantly, may break the dynamic-programming structure. As observed by Fei et al. (2021), losing a Bellman-type recursion can cause error terms to compound over the horizon, leading to substantially worse dependence on H in the order of $e^{\beta H^2}$. Following Fei et al. (2021), we instead work directly in the exponential space induced by the criterion. This restores a Bellman recursion for the exponentiated value functions and enables sharper control of uncertainty; in particular, 20 derives Bellman-type variance identities in this space, which are unavailable in the original entropic-value space due to the lack of sub-linearity. This identity highlights that the exponential parameterization is the natural domain for controlling uncertainty under the entropic criterion.

For a policy π , recall the entropic value functions $V_h^\pi(s) = \frac{1}{\beta} \log \mathbb{E}[\exp(\beta \sum_{h'=h}^H r_{h'}(s_{h'}, a_{h'})) \mid s_h = s]$. The exponential transforms of the value and the state-value function are denoted as:

$$Z_h^\pi(s) \triangleq \exp(\beta V_h^\pi(s)), \quad U_h^\pi(s, a) \triangleq \exp(\beta Q_h^\pi(s, a)).$$

In this section, we represent our algorithm (Name) along with a high probability upper bound on its sampling complexity. The algorithm is a UCB-style approach that maintains a KL-confidence sets around the unknown transition model defined as:

$$\mathcal{C}_h^t(s, a) \triangleq \left\{ q \in \Sigma_S : \text{KL}(\hat{p}_h^t(s, a), q(s, a)) \leq \frac{\alpha(n_h^t(s, a), \delta)}{n_h^t(s, a)} \right\}$$

The algorithm then plans optimistically over all models in these sets, using dynamic programming in the exponential space. The sign of β changes the direction of optimism. We present the construction for $\beta > 0$; the $\beta < 0$ case follows by symmetric modifications Appendix B.2. Full proofs are deferred to Appendix B.

The novelty of our approach is twofold. We build KL-based exploration bonuses that are compatible with the entropic criterion and admit variance-sensitive control in exponential space. We also propose an entropic stopping rule that yields sharper horizon dependence, improving over bounds that incur an additional factor of order $e^{|\beta|H}$ (Mortensen and Talebi, 2025), (Fei et al., 2021).

Similar Azar et al. (2017), Zanette and Brunskill (2019) and Ménard et al. (2021) we define optimistic and pessimistic state-value function on the exponential transform of Q_h^* which denoted by U_h^* : The optimistic value function is defined by:

$$\begin{aligned} \tilde{U}_h^t(s, a) &= \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \tilde{Z}_h^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathcal{Z}_{h+1}^t)(s, a) \right] \right\} \\ \tilde{Z}_h^t(s) &= \max_{a \in \mathcal{A}} \tilde{U}_h^t(s, a) \\ \tilde{Z}_{H+1}^t(s) &= 0 \end{aligned}$$

Where the bonus term is defined as:

$$b_h^t(s, a) = 2\sqrt{2} \sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 5e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + 4He^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \quad (1)$$

α^* is an exploration rate that comes from the Bernstein inequality. As stated before, our algorithm acts greedily on this optimistic value function, we thus define our policy as:

$$\tilde{U}_h^t(s, a) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \tilde{Z}_h^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathcal{Z}_{h+1}^t)(s, a) \right] \right\}$$

Stopping rule via an entropic certificate. Next, we define a certificate $G_h^t : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}_+$ computed by a backward recursion (alongside the optimistic/pessimistic estimates) :

$$\pi_{h+1}^{t+1} G_h^t(s) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h(s, \pi_h^{t+1}(s))} \right\} 2b_h^t(s, \pi_h^{t+1}(s)) + \left(1 + \frac{2}{H} \right) \tilde{p}_h^t(\pi_{h+1}^{t+1} G_{h+1}^t)(s, \pi_h^{t+1}(s))$$

with terminal $G_{H+1}^t \equiv 0$. In particular, on the high-probability good event, the performance gap of the greedy policy π_{t+1} is controlled by the start-state certificate. Concretely, Lemma Theorem 9 shows that

$$Z_1^*(s_1) - Z_1^{\pi_{t+1}}(s_1) \leq (\pi_{t+1,1} G_1^t)(s_1),$$

Stopping rule We next define a stopping rule based on the certificate G_h^t by going back from the exponential space to ... we write:

$$(V_1^\star - V_1^{\pi^{t+1}})(s_1) = \frac{1}{\beta} \log \left(\frac{Z_1^\star}{Z_1^{\pi^{t+1}}} \right)(s_1) = \frac{1}{\beta} \log \left(1 + \frac{Z_1^\star - Z_1^{\pi^{t+1}}}{Z_1^{\pi^{t+1}}} \right)(s_1)$$

But since $Z_1^{\pi^{t+1}} - Z_1^\star \leq \pi_1 G_1(s)$ then:

$$(V_1^\star - V_1^{\pi^{t+1}})(s_1) \leq \frac{1}{\beta} \log \left(1 - \frac{\pi_1 G_1(s)}{Z_1^{\pi^{t+1}}(s)} \right)$$

Hence to have the policy be ε -optimal, we must have:

$$\pi_1^\tau G_1(s_1) \leq (1 - e^{\beta\varepsilon}) Z_1^{\pi^\tau}$$

$$(V_1^\star - V_1^{\pi^{t+1}})(s_1) = \frac{1}{\beta} \log \left(\frac{Z_1^\star}{Z_1^{\pi^{t+1}}} \right)(s_1) = \frac{1}{\beta} \log \left(1 + \frac{Z_1^\star - Z_1^{\pi^{t+1}}}{Z_1^{\pi^{t+1}}} \right)(s_1)$$

But since $Z_1^{\pi^{t+1}} - Z_1^\star \leq \pi_1 G_1(s)$ then:

$$(V_1^\star - V_1^{\pi^{t+1}})(s_1) \leq \frac{1}{\beta} \log \left(1 + \frac{\pi_1 G_1(s)}{Z_1^{\pi^{t+1}}} \right)(s_1)$$

Hence to have the policy be ε -optimal at time τ , we must have:

$$\pi_1^\tau G_1(s_1) \leq (e^{\beta\varepsilon} - 1) Z_1^{\pi^\tau}$$

We are now ready to write the algorithm : We prove a sampling complexity bound for (algorithm name) in the following theorem. This complexity bound is valid for both $\beta > 0$ (proof in appendix B.1) and for $\beta < 0$ (proof in appendix B.2)

Theorem 3 (Sampling complexity) For $\delta \in [0, 1]$ and $\varepsilon \in]0, \frac{2}{|\beta|HS}]$, the algorithm (name) is (ε, δ) -PAC for best policy identification problem for the entropic risk measure after τ episodes. Moreover, with probability $1 - \delta$:

$$\tau \leq \frac{(e^{|\beta|H} - 1)HSA}{(e^{\beta\varepsilon} - 1)^2} \log\left(\frac{4SAH}{\delta}\right) C_2^2$$

Where $C_2 = 2765e^{22} \log \left(4e^{17} \frac{(S+1)(H+1)e^{|\beta|H} SAH^2}{e^{\beta\varepsilon} - 1} \right)$. In particular, hiding constants and log terms:

$$\tau = \tilde{O} \left(\frac{(e^{|\beta|H} - 1)HSA}{(e^{|\beta|\varepsilon} - 1)^2} \right)$$

Proof [Sketch of proof] We first prove that our $\tilde{U}_h^t$ and $\underline{U}_h^t$ are indeed optimistic and pessimistic towards the exponential transform of the value function $U_h^\star$ in the optimism lemma 6. Then similarly

Algorithm 1 Entropic-BPI (single backward loop over h)

- 1: **Input:** $\beta > 0, \delta \in (0, 1), \varepsilon > 0$.
 - 2: Initialize counts $n_h^0(\cdot) = 0$ and $\hat{p}_h^0(\cdot|s, a) = 1/S$.
 - 3: **for** $t = 0, 1, 2, \dots$ **do**
 - 4: **Terminal init:** set $\tilde{Z}_{H+1}^t(s) = 1, G_{H+1}^t(s) = 0$, and $Z_{H+1}^{\pi^{t+1}}(s) = 1$ for all s .
 - 5: **for** $h = H, H-1, \dots, 1$ **do**
 - 6: Compute bonus $\tilde{b}_h^t(\cdot)$ (1).
 - 7: **Optimistic backup:** for all (s, a) compute $\tilde{U}_h^t(s, a)$ using $\tilde{Z}_{h+1}^t$
 - 8: Set $\tilde{Z}_h^t(s) = \max_{a \in \mathcal{A}} \tilde{U}_h^t(s, a)$ for all s .
 - 9: **Sampling rule:** $\pi_h^{t+1}(s) \in \arg \max_{a \in \mathcal{A}} \tilde{U}_h^t(s, a) \quad \forall s$.
 - 10: **Certificate (backward recursion):** for all s compute
$$\pi_{h+1}^{t+1} G_h^t(s) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h^t(s, \pi_h^{t+1}(s))} \left[2b_h^t(s, \pi_h^{t+1}(s)) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t \pi_{h+1}^{t+1} G_{h+1}^t(s) \right] \right\}.$$
 - 11: **end for**
 - 12: **Stopping test (after finishing the backward loop):**
 - 13: **if** $(\pi_1^{t+1} G_1^t)(s_1) \leq (e^{\beta \varepsilon} - 1) Z_1^{\pi^{t+1}}(s_1)$ **then**
 - 14: **stop** and output π^{t+1} .
 - 15: **end if**
 - 16: Execute episode $t+1$ with π^{t+1} , update counts and $\hat{p}_h^{t+1}$.
 - 17: **end for**
-

to (Ménard et al., 2021) (Dann et al., 2017) we proceed to bound the width certificate by a computable recursive upper bound that it is used as a stopping rule for our algorithm. More precisely, by defining the width certificate as :

$$\pi_{h+1}^{t+1} G_h^t(s) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h^t(s, \pi_h^{t+1}(s))} \right\} 2b_h^t(s, \pi_h^{t+1}(s)) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t (\pi_{h+1}^{t+1} G_{h+1}^t)(s, \pi_h^{t+1}(s))$$

And lemma 9 shows that with good probability this width certificate upper bound the optimality gap:

$$Z_1^*(s_1) - Z_1^{\pi^{t+1}}(s_1) \leq \pi_{t+1,1} G_1^t(s_1)$$

Now we unroll the recursive equation on $\pi_{h+1}^t G_h^t$ to derive when the sampling complexity is verified:

$$\begin{aligned} (\pi_1 G_1^t)(s_1) &\leq \mathbb{E}^\pi \left[\sum_{h=1}^H \kappa^{h-1} \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(12 \sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \alpha^*(n_h^t(s_h, a_h) \wedge 1)} \right. \right. \\ &\quad \left. \left. + 63 H e^{\beta(H-h)} \alpha(n_h^t(s_h, a_h) \wedge 1) \right) \right] \Big|_{s_1} \end{aligned}$$

Now, we write the stopping rule:

$$(V_1^* - V_1^{\pi^{t+1}})(s_1) = \frac{1}{\beta} \log \left(\frac{Z_1^*}{Z_1^{\pi^{t+1}}} \right)(s_1) = \frac{1}{\beta} \log \left(1 + \frac{Z_1^* - Z_1^{\pi^{t+1}}}{Z_1^{\pi^{t+1}}} \right)(s_1) \leq \frac{1}{\beta} \frac{(Z_1^* - Z_1^{\pi^{t+1}})(s_1)}{Z_1^{\pi^{t+1}}(s_1)} \leq \frac{1}{\beta} \frac{\pi_1 G_1(s_1)}{Z_1^{\pi^{t+1}}}$$

Hence to have the policy be ε -optimal at time τ , we must have:

$$\pi_1^\tau G_1(s_1) \leq (e^{\beta\varepsilon} - 1)Z_1^{\pi^\tau}$$

But

$$\begin{aligned} & \frac{1}{Z_1^{\pi^{t+1}}}(\pi_1 G_1^t)(s_1) \leq \\ & e^{11} \mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(12 \sqrt{\frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \alpha(n_h^t(s_h, a_h) \wedge 1) \right. \right. \\ & \quad \left. \left. + 63H e^{\beta(H-h)} \alpha(n_h^t(s_h, a_h) \wedge 1) \right) \middle| s_1 \right] \end{aligned}$$

We bound the first term using Cauchy-Schwartz inequality:

$$\begin{aligned} & \mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(\sqrt{\frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \left(\frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \right) \middle| s_1 \right] \\ & \leq \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp \left(2\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \end{aligned}$$

By lemma 20 we have:

$$\sum_{h=1}^H \mathbb{E}^\pi \left[e^{2\beta \sum_{i=1}^h r_i(s_i, a_i)} \frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2} \right] = \mathbb{E}^\pi \left[\frac{\sigma V_1^\pi(s_1)}{(Z_1^\pi)^2} \right]$$

And then using lemma 21 we find:

$$\frac{\sigma V_1^\pi(s_1)}{(Z_1^\pi)^2} \leq e^{\beta H} - 1$$

For the second term:

$$\begin{aligned} & \mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) H e^{\beta(H-h)} \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \middle| s_1 \right] \\ & = \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) H e^{\beta(H-h)} \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \\ & \leq \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) e^{\beta h} H e^{\beta(H-h)} \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \\ & \leq H e^{\beta H} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \end{aligned}$$

■

4. Experiments

5. Discussions and Conclusions

Discuss that this closes a gap but opens questions as the overall complexity is exponential and so clearly not friendly. In particular not friendly to extensions to VaR and CVaR as in Marthe et al. So the question of feasible MDPs, and friendlier structure is open and should be studied.

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Appendix

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Appendix A. Concentration events

Following the ideas of (Ménard et al., 2021) we define the following quantities:

$$\begin{aligned}\beta(n, \delta) &\triangleq \log(4SAH/\delta) + S \log(8e(n+1)) \\ \beta^{\text{cnt}}(\delta) &\triangleq \log(4SAH/\delta) \quad \text{and} \\ \beta^*(n, \delta) &\triangleq \log(4SAH/\delta) + \log(8e(n+1))\end{aligned}$$

We also define the KL-divergence concentration event as:

$$\mathcal{E}_{KL} = \left\{ \forall t \in \mathbb{N}, \forall h \in [H], \forall (s, a) \in \mathcal{S} \times \mathcal{A} : D_{KL}(\tilde{p}_h^t(s, a), p_h(s, a)) \leq \alpha(n_h^t(s, a), \delta) \right\}$$

and the Bernoulli concentration event for a series of function $(f_h)_{h \in [H+1]}$:

$$\mathcal{E}_f = \left\{ \forall t \in \mathbb{N}, \forall h \in [H], \forall (s, a) \in \mathcal{S} \times \mathcal{A} : \right. \\ \left. |(\hat{p}_h^t - p_h)f_{h+1}(s, a)| < \sqrt{2 \text{Var}_{p_h}(f_{h+1})(s, a) \frac{\alpha^*(n_h^t(s, a), \delta)}{n_h^t(s, a)}} + 3\mathbb{S}(f_{h+1}) \frac{\alpha^*(n_h^t(s, a), \delta)}{n_h^t(s, a)} \right\}$$

And the counts concentration event:

$$\mathcal{E}^{\text{cnt}} = \left\{ \forall t \in \mathbb{N}, \forall h \in [H], \forall (s, a) \in \mathcal{S} \times \mathcal{A} : n_h^t(s, a) \geq \frac{1}{2} \bar{n}_h^t(s, a) - \alpha^{\text{cnt}}(\delta) \right\}$$

Using this, we define the good event for our algorithm analysis for $\beta > 0$ and $\beta < 0$: for $\beta > 0$

$$\mathcal{E}^+ = \mathcal{E}_{KL} \cap \mathcal{E}_{V^*} \cap \mathcal{E}_{\hat{V}} \cap \mathcal{E}^{\text{cnt}}$$

And for $\beta < 0$ we have almost the same thing but the definition of $\hat{V}$ and the range of the functions changes:

$$\mathcal{E}^- = \mathcal{E}_{KL} \cap \mathcal{E}_{V^*} \cap \mathcal{E}_{\hat{V}} \cap \mathcal{E}^{\text{cnt}}$$

Lemma 4 *It holds that :*

$$\mathbb{P}(\mathcal{E}_{KL}) \geq 1 - \frac{\delta}{4}, \quad \mathbb{P}(\mathcal{E}^{\text{cnt}}) \geq 1 - \frac{\delta}{4}, \quad \text{and} \quad \text{for any } f \quad \mathbb{P}(\mathcal{E}_f) \geq 1 - \frac{\delta}{4}$$

Consequently,

$$\mathbb{P}(\mathcal{E}^+) \geq 1 - \delta \quad \text{and} \quad \mathbb{P}(\mathcal{E}^-) \geq 1 - \delta$$

Proof The KL concentration event:

For (h, s, a) fixed, we apply lemma 16 with confidence level $\delta_{h,s,a} = \frac{\delta}{4SAH}$ and then do a union bound over h, s, a to get a concentration inequality that holds uniformly over $[H] \times \mathcal{S} \times \mathcal{A}$ **The Bernstein concentration event:**

Let $(f_h)_{h \in [H+1]}$ be a sequence of function. Fix (h, s, a) and let $(\mathcal{F}_\tau)_{\tau \geq 0}$ be the history filtration. Define

$$w_\tau := \mathbf{1}\{(H_\tau, S_\tau, A_\tau) = (h, s, a)\}, \quad Y_\tau := f_{h+1}(S_{\tau+1}) - \mathbb{E}[f_{h+1}(S_{\tau+1}) \mid \mathcal{F}_{\tau-1}]$$

Then (w_τ) is predictable, $\mathbb{E}[Y_\tau \mid \mathcal{F}_{\tau-1}] = 0$, $|Y_\tau| \leq H$, and

$$\mathbb{E}[Y_\tau^2 \mid \mathcal{F}_{\tau-1}] = \text{Var}_{p_h}(f_{h+1})(s, a) \quad \text{on } \{w_\tau = 1\}$$

Let

$$S_t = \sum_{\tau=1}^t w_\tau Y_\tau, \quad V_t = \sum_{\tau=1}^t w_\tau^2 \mathbb{E}[Y_\tau^2 \mid \mathcal{F}_{\tau-1}], \quad W_t = \sum_{\tau=1}^t w_\tau = n_h^t(s, a)$$

Then $V_t = W_t \text{Var}_{p_h}(f_{h+1})(s, a)$ and, for $W_t \geq 1$,

$$(\hat{p}_h^t - p_h)f_{h+1}(s, a) = \frac{S_t}{W_t}$$

Applying lemma 19 with $b = \mathbb{S}(f)$ and confidence $\delta_{h,s,a}$ yields (simultaneously for all t)

$$\left| (\hat{p}_h^t - p_h)f_{h+1}(s, a) \right| \leq \sqrt{2 \text{Var}_{p_h}(f_{h+1})(s, a) \frac{\alpha^*(n_h^t(s, a), \delta)}{n_h^t(s, a)}} + 3\mathbb{S}(f) \frac{\alpha^*(n_h^t(s, a), \delta)}{n_h^t(s, a)}$$

where $\alpha^*(n, \delta) := \log\left(\frac{4e(2n+1)}{\delta_{h,s,a}}\right)$ (using $n_h^t(s, a) \leq t$ and monotonicity of the log term). Finally choose $\delta_{h,s,a} = \frac{\delta}{HSA}$ and we apply a union bound over (h, s, a) to get the result for any state-action pair

The counts concentration event:

The proof follows from lemma 17 applied to the Bernoulli random variable $\mathbb{1}\{(s_h^i, a_h^i) = (s, a)\}$ for $\delta_{h,s,a} = \frac{\delta}{HSA}$ and then doing a union bound over h, s, a ■

Appendix B. Algorithm analysis

Here we provide a detailed analysis of our algorithm. Our method is a UCB-style algorithm that plans over a KL confidence region, following the approach of Ménard et al. (2021) for the risk-neutral objective. At each step t and stage h , we construct a confidence set around the true transition kernel:

$$\mathcal{C}_h^t(s, a) \triangleq \left\{ q \in \Sigma_S : \text{KL}(\hat{p}_h^t(s, a), q(s, a)) \leq \frac{\alpha(n_h^t(s, a), \delta)}{n_h^t(s, a)} \right\}$$

The algorithm then acts optimistically by selecting, among all transition models q such that $q(\cdot | s, a) \in \mathcal{C}_h^t(s, a)$, the one that yields the highest value function, and plans accordingly.

For the entropic risk measure, we follow the same principle, but carry out optimistic planning in the exponential (log-moment-generating) space induced by the entropic criterion. As noted by Fei et al. (2021), working in this exponential space allows us to exploit a Bellman-type recursion that is generally lost if one applies Lipschitz arguments directly in the original value space. This means that the upper and lower confidence bounds on the optimal exponential transformation of the

state-value function U^* and value function Z^* for $\beta > 0$ are given by:

$$\begin{aligned}
 \bar{U}_h^t(s, a) &\triangleq e^{\beta r_h(s, a)} \max_{\bar{p}_h \in \mathcal{C}_h^t(s, a)} \bar{p}_h \bar{Z}_{h+1}^t(s, a) & \underline{U}_h^t(s, a) &\triangleq e^{\beta r_h(s, a)} \min_{\underline{p}_h \in \mathcal{C}_h^t(s, a)} \underline{p}_h \underline{Z}_{h+1}^t(s, a) \\
 \bar{Z}_h^t(s) &\triangleq \max_a \bar{U}_h^t(s, a) & \underline{Z}_h^t(s) &\triangleq \max_a \underline{U}_h^t(s, a) \\
 \bar{Z}_{H+1}^t(s) &\triangleq 0 & \underline{Z}_{H+1}^t(s) &\triangleq 0 \\
 \bar{p}_h^t(s, a) &\in \arg \max_{\bar{p} \in \mathcal{C}_h^t(s, a)} \bar{p}_h \bar{Z}_{h+1}^t(s, a) & \underline{p}_h^t(s, a) &\in \arg \min_{\underline{p} \in \mathcal{C}_h^t(s, a)} \underline{p}_h \underline{Z}_{h+1}^t(s, a) \\
 \bar{\pi}_h^t(s, a) &\in \arg \max_{a \in \mathcal{A}} \bar{U}_h^t(s, a) & \underline{\pi}_h^t(s, a) &\in \arg \max_{a \in \mathcal{A}} \underline{U}_h^t(s, a).
 \end{aligned}$$

For $\beta < 0$, $\bar{U}$ will correspond to the pessimistic $\underline{Q}$ via the log-transformation. As such, maximizing $\bar{Q}$ to define the policy $\bar{\pi}$ is equivalent to minimizing $\underline{U}$. Similarly, finding the best action at each stage to define $\bar{Z}$ and $\underline{Z}$ corresponds to minimizing $\bar{U}$ and $\underline{U}$ respectively:

$$\begin{aligned}
 \bar{U}_h^t(s, a) &\triangleq e^{\beta r_h(s, a)} \min_{\bar{p}_h \in \mathcal{C}_h^t(s, a)} \bar{p}_h \bar{Z}_{h+1}^t(s, a) & \underline{U}_h^t(s, a) &\triangleq e^{\beta r_h(s, a)} \max_{\underline{p}_h \in \mathcal{C}_h^t(s, a)} \underline{p}_h \underline{Z}_{h+1}^t(s, a) \\
 \bar{Z}_h^t(s) &\triangleq \min_a \bar{U}_h^t(s, a) & \underline{Z}_h^t(s) &\triangleq \min_a \underline{U}_h^t(s, a) \\
 \bar{Z}_{H+1}^t(s) &\triangleq 0 & \underline{Z}_{H+1}^t(s) &\triangleq 0 \\
 \bar{p}_h^t(s, a) &\in \arg \min_{\bar{p} \in \mathcal{C}_h^t(s, a)} \bar{p}_h \bar{Z}_{h+1}^t(s, a) & \underline{p}_h^t(s, a) &\in \arg \max_{\underline{p} \in \mathcal{C}_h^t(s, a)} \underline{p}_h \underline{Z}_{h+1}^t(s, a) \\
 \bar{\pi}_h^t(s, a) &\in \arg \min_{a \in \mathcal{A}} \bar{U}_h^t(s, a) & \underline{\pi}_h^t(s, a) &\in \arg \max_{a \in \mathcal{A}} \underline{U}_h^t(s, a).
 \end{aligned}$$

We now proceed to turn these definitions into computable optimistic and pessimistic expressions using only the empirical MDP. We then prove the corresponding optimism lemma, bound the certificate width, and derive the sample complexity. We treat the cases $\beta > 0$ and $\beta < 0$ separately.

B.1. Case $\beta > 0$

Let us start with a concentration inequality:

Lemma 5 *On the good event $\mathcal{E}^+$ we have:*

$$\begin{aligned}
 |(p_h - \hat{p}_h^t) Z_{h+1}^*(s, a)| &\leq 2\sqrt{2} \sqrt{\text{Var}_{\hat{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} + 5e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + 4He^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\
 &\quad + \frac{1}{H} \hat{p}_h^t (\tilde{Z}_{h+1}^t - Z_{h+1}^*)(s, a)
 \end{aligned}$$

Proof On the good event $\mathcal{E}^+$ we have:

$$|(p_h - \hat{p}_h^t) Z_{h+1}^*(s, a)| \leq \sqrt{2 \text{Var}_{p_h}(Z_{h+1}^*) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 3e^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}$$

We apply lemma 22 and lemma 23 successively to transport the variance of Z_{h+1}^* under p_h to the computable variance of $\tilde{Z}_{h+1}^t$ under $\tilde{p}_h^t$

$$\begin{aligned}\text{Var}_{p_h}(Z_{h+1}^*) &\leq 2 \text{Var}_{\tilde{p}_h^t}(Z_{h+1}^*)(s, a) + 4e^{2\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\ &\leq 4 \text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t)(s, a) + 4e^{\beta(H-h)} \tilde{p}_h^t(\tilde{Z}_{h+1}^t - Z_{h+1}^*)(s, a) + 4e^{2\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}\end{aligned}$$

Hence, using the inequality $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ and then $\sqrt{ab} \leq a\eta + \frac{b}{\eta}$ for $\eta = H$ and using that $\alpha^*(n, \delta) \leq \alpha(n, \delta)$:

$$\begin{aligned}\sqrt{\text{Var}_{p_h}(Z_{h+1}^*) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} &\leq 2\sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + \sqrt{4e^{\beta(H-h)} \tilde{p}_h^t(\tilde{Z}_{h+1}^t - Z_{h+1}^*)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \\ &\quad + 2e^{\beta(H-h)} \sqrt{\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \\ &\leq 2\sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4He^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\ &\quad + \frac{1}{H} \tilde{p}_h^t(\tilde{Z}_{h+1}^t - Z_{h+1}^*)(s, a) + 2e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}\end{aligned}$$

Hence, using again that $\alpha^*(n, \delta) \leq \alpha(n, \delta)$ we obtain:

$$\begin{aligned}|(p_h - \tilde{p}_h^t) Z_{h+1}^*| &\leq 2\sqrt{2} \sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 5e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + 4He^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\ &\quad + \frac{1}{H} \tilde{p}_h^t(\tilde{Z}_{h+1}^t - Z_{h+1}^*)(s, a)\end{aligned}$$

■

Denote the bonus term:

$$b_h^t(s, a) = 2\sqrt{2} \sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 5e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + 4He^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}$$

Now, following Azar et al. (2017), Zanette and Brunskill (2019) and Ménard et al. (2021) we define optimistic and pessimistic state-value function on the exponential transform of Q_h^* which denoted by U_h^* :

$$\tilde{U}_h^t(s, a) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \tilde{Z}_h^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t(\tilde{Z}_{h+1}^t - \tilde{Z}_{h+1}^t)(s, a) \right] \right\}$$

And

$$\tilde{Z}_h^t(s) = \max_{a \in \mathcal{A}} \tilde{U}_h^t(s, a)$$

$$\tilde{Z}_{H+1}^t(s) = 0$$

We consider the greedy policy:

$$\pi_h^{t+1}(s) = \arg \max_{a \in \mathcal{A}} \tilde{U}_h^t(s, a)$$

And a pessimistic lower bound

$$\underline{U}_h^t(s, a) = \max \left\{ 1, e^{\beta r_h(s, a)} \left[\tilde{p}_h^t Z_h^t(s, a) - b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - Z_{h+1}^t)(s, a) \right] \right\}$$

And

$$\underline{Z}_h^t(s) = \max_{a \in \mathcal{A}} \underline{U}_h^t(s, a)$$

$$\underline{Z}_{H+1}^t(s) = 0$$

Now let us prove the optimism lemma:

Lemma 6 *With high probability $1 - \delta$ we have:*

$$\underline{U}_h^t(s, a) \leq U_h^*(s, a) \leq \tilde{U}_h^t(s, a)$$

and

$$\underline{Z}_h^t(s) \leq Z_h^*(s) \leq \tilde{Z}_h^t(s)$$

Proof We proceed by induction over h . For $h = H + 1$ the result is trivially upper bounding and (resp. lower bounding) Q^* by H and 1.

Assume the inequality holds for $h' > h$. Fix (s, a) and assume $\tilde{Q}_h^t(s, a) < H$. we have that:

$$\begin{aligned} \tilde{U}_h^t(s, a) - U_h^*(s, a) &= e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - Z_{h+1}^t)(s, a) - p_h Z_{h+1}^*(s, a) \right] \\ &= e^{\beta r_h(s, a)} \left[\tilde{p}_h^t (\tilde{Z}_{h+1}^t(s, a) - Z_{h+1}^*(s, a)) + (\tilde{p}_h^t - p_h) Z_{h+1}^*(s, a) \right. \\ &\quad \left. + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - Z_{h+1}^t)(s, a) \right] \end{aligned}$$

But we know by Bernstein inequality that:

$$(\tilde{p}_h^t - p_h) Z_{h+1}^*(s, a) \geq -b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - Z_{h+1}^*(s, a))$$

Hence:

$$\tilde{U}_h^t(s, a) - U_h^*(s, a) \geq e^{\beta r_h(s, a)} \left[\left(1 - \frac{1}{H}\right) \tilde{p}_h^t (\tilde{Z}_{h+1}^t(s, a) - Z_{h+1}^*(s, a)) + \frac{1}{H} \tilde{p}_h^t (Z_{h+1}^*(s, a) - Z_{h+1}^t(s, a)) \right] \geq 0$$

Where we used the induction hypothesis. We prove the pessimistic property in the same way ■

B.1.1. STOPPING RULE

We define the stopping rule for the algorithm for the case $\beta > 0$:

$$G_h^t(s, a) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \hat{p}_h^t \pi^{t+1} G_{h+1}^t(s) \right] \right\}$$

Lemma 7 establishes the validity of this stopping rule by showing that, with high probability, it bounds the certificate width:

Lemma 7 *On the good event $\mathcal{E}^+$, for all t and all h ,*

$$Z_h^*(s) - Z_h^{\pi^{t+1}}(s) \leq \pi_h^{t+1} G_h^t(s) \quad \forall s \in \mathcal{S}$$

In particular, at the initial state s_1 , $Z_1^(s_1) - Z_1^{\pi^{t+1}}(s_1) \leq \pi_1^{t+1} G_1^t(s_1)$*

We prove the lemma 7 in this section :

We introduce the “ring” quantities $\hat{U}_h^t$ and $\hat{Z}_h^t$ as auxiliary, analysis-only value functions. These functions are constructed to satisfy the exponential Bellman recursion under the true transition kernel p_h , while additionally being constrained to lie below a pessimistic empirical backup. By construction, $\hat{Z}_h^t$ is dominated both by the algorithm’s pessimistic estimate and by the exponentiated value of the greedy policy π^{t+1} .

Since $\tilde{Z}$ is defined to be pessimistic with respect to Z^* , it does not directly yield a comparison to the value of an arbitrary policy V^π . The ring quantity $\hat{Z}_h^t$ resolves this mismatch by serving as a common lower bound—a worst-case proxy that lies below both $\tilde{Z}_h^t$ and $Z_h^{\pi^{t+1}}$:

- Terminal: $\hat{Z}_{H+1}^t(s) = 1$ for all s .
- For $h = H, H-1, \dots, 1$:

1. Define the “true-backup candidate”

$$\hat{U}_{h,\text{true}}^t(s, a) := e^{\beta r_h(s, a)} (p_h \hat{Z}_{h+1}^t)(s, a)$$

2. Define a “pessimistic-backup candidate” $\hat{U}_{h,\text{pes}}^t(s, a)$ by taking the same functional form as the pessimistic estimator, but plugging $\hat{Z}_{h+1}^t$ wherever the estimator uses a next-step value input:

$$\hat{U}_{h,\text{pes}}^t(s, a) := \max \left\{ 1, e^{\beta r_h(s, a)} \left[(\hat{p}_h^t \hat{Z}_{h+1}^t)(s, a) - b_h^t(s, a) - \frac{1}{H} \hat{p}_h^t (\tilde{Z}_{h+1}^t - \hat{Z}_{h+1}^t)(s, a) \right] \right\}$$

3. Then set

$$\hat{U}_h^t(s, a) := \min \left\{ \hat{U}_{h,\text{true}}^t(s, a), \hat{U}_{h,\text{pes}}^t(s, a) \right\}$$

4. Finally define the ring value along the greedy policy:

$$\hat{Z}_h^t(s) := \hat{U}_h^t(s, \pi_h^{t+1}(s))$$

We have the following:

Lemma 8 For all (h, s, a) :

$$\mathring{U}_h^t(s, a) \leq \min \left(\mathring{U}_h^t(s, a), U_h^{\pi_h^{t+1}}(s, a) \right)$$

and

$$\mathring{Z}_h^t(s) \leq \min \left(\mathring{Z}_h^t(s), Z_h^{\pi_h^{t+1}}(s, a) \right)$$

Proof We proceed by backward induction. For $h = H + 1$, all values are equal to 0 so the inequalities hold. Assume that for some $h \leq H$ we have for all (s, a) :

$$\mathring{U}_{h+1}^t(s, a) \leq \min \left(\mathring{U}_{h+1}^t(s, a), U_h^{\pi_{h+1}^{t+1}}(s, a) \right)$$

and

$$\mathring{Z}_{h+1}^t(s) \leq \min \left(\mathring{Z}_{h+1}^t(s), Z_h^{\pi_{h+1}^{t+1}}(s, a) \right)$$

we have by construction:

$$\mathring{U}_h^t(s, a) \leq \mathring{U}_{h,\text{true}}^t(s, a) = e^{\beta r_h(s, a)} (p_h \mathring{Z}_{h+1}^t)(s, a) \leq e^{\beta r_h(s, a)} (p_h Z_{h+1}^{\pi_{h+1}^{t+1}})(s, a) = U_h^{\pi_{h+1}^{t+1}}(s, a)$$

Where we used the induction hypothesis and the monotonicity of the exponential Bellman operator. Again by construction;

$$\mathring{U}_h^t(s, a) \leq \mathring{U}_{h,\text{pes}}^t(s, a)$$

But since we have:

$$\begin{aligned} \mathring{U}_h^t(s, a) - \mathring{U}_{h,\text{pes}}^t(s, a) &= e^{\beta r_h(s, a)} \left[\left(\tilde{p}_h^t \mathring{Z}_{h+1}^t(s, a) - b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathring{Z}_{h+1}^t)(s, a) \right) \right. \\ &\quad \left. - \left(\tilde{p}_h^t \mathring{Z}_{h+1}^t(s, a) - b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathring{Z}_{h+1}^t)(s, a) \right) \right] \\ &= e^{\beta r_h(s, a)} \left(1 - \frac{1}{H} \right) \left[\tilde{p}_h^t (\mathring{Z}_{h+1}^t - \mathring{Z}_{h+1}^t) \right] \\ &\geq 0 \end{aligned}$$

Where we applied the induction hypothesis, we conclude then:

$$\mathring{U}_h^t(s, a) \leq \mathring{U}_h^t(s, a)$$

The bound on V follows immediately and we conclude the recursion ■

On the good event $\mathcal{E}^+$ we have:

Lemma 9 On the good event $\mathcal{E}^+$

$$\tilde{U}_h^t(s, a) - \mathring{U}_h^t(s, a) \leq e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathring{Z}_{h+1}^t) \right]$$

Proof Fix a state-action pair (s, a) and $h \in [H]$, we consider two cases:

First case: $\hat{U}_h^t(s, a) = \hat{U}_{h, \text{true}}^t(s, a)$ we then have:

$$\tilde{U}_h^t(s, a) - \hat{U}_h^t(s, a) \leq e^{\beta r_h^t(s, a)} \left[b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t \right) (s, a) + \tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) - p_h \dot{Z}_{h+1}^t(s, a) \right]$$

The last term can be written as:

$$\tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) - p_h \dot{Z}_{h+1}^t(s, a) = \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \dot{Z}_h^t \right) (s, a) + \left(\tilde{p}_h^t - p_h \right) \dot{Z}_h^t(s, a)$$

For the second term, since $\mathcal{E}_{\hat{V}} \subset \mathcal{E}^+$ then on the event $\mathcal{E}^+$:

$$\left| \left(\tilde{p}_h^t - p_h \right) \dot{Z}_h^t(s, a) \right| \leq 2 \sqrt{\text{Var}_{p_h}(\dot{Z}_h^t) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} + \frac{2}{3} e^{\beta H} \alpha(n_h^t(s, a))$$

Next, we use the variance transportation inequality:

$$\sqrt{\text{Var}_{p_h}(\dot{Z}_{h+1}^t) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} \leq 2 \sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} + 4H e^{\beta H} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t)$$

Hence:

$$\left| \left(\tilde{p}_h^t - p_h \right) \dot{Z}_h^t(s, a) \right| \leq b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t \right)$$

Hence by combining the two bounds:

$$\tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) - p_h \dot{Z}_{h+1}^t(s, a) \leq b_h^t(s, a) + \left(1 + \frac{1}{H} \right) \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t \right)$$

Hence by substituting and using lemma 8:

$$\tilde{U}_h^t(s, a) - \hat{U}_h^t(s, a) \leq e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t \right) \right]$$

Second case: $\hat{U}_h^t(s, a) = \hat{U}_{h, \text{pes}}^t(s, a)$. In this case:

$$\begin{aligned} \tilde{U}_h^t(s, a) - \hat{U}_h^t(s, a) &\leq e^{\beta r_h^t(s, a)} \left[b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t) (s, a) + \tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) \right. \\ &\quad \left. - \left(\tilde{p}_h^t \dot{Z}_{h+1}^t(s, a) - b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t) (s, a) \right) \right] \\ &= e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{1}{H} \right) \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t) + \frac{1}{H} (\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t) \right] (s, a) \end{aligned}$$

Hence by lemma 8 we find:

$$\tilde{U}_h^t(s, a) - \hat{U}_h^t(s, a) \leq e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t \right) \right]$$

Which conclude the recursion ■

We now prove lemma 7:

Proof We first prove by backward induction that, for all h and s ,

$$\tilde{Z}_h^t(s) - \dot{Z}_h^t(s) \leq (\pi_h^{t+1} G_h^t)(s). \quad (2)$$

For $h = H + 1$ it holds since both sides are 0. Assume it holds at step $h + 1$. For $a = \pi_h^{t+1}(s)$

$$\tilde{Z}_h^t(s) - \dot{Z}_h^t(s) = \tilde{U}_h^t(s, a) - \dot{U}_h^t(s, a).$$

By lemma 9, we have :

$$\tilde{U}_h^t(s, a) - \dot{U}_h^t(s, a) \leq e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t \right) \right]$$

By the induction hypothesis inside the tilted expectation:

$$\tilde{p}_h^t(\tilde{Z}_{h+1}^t - \dot{Z}_{h+1}^t)(s, a) \leq \tilde{p}_h^t(\pi^{t+1} G_{h+1}^t)(s, a).$$

Thus,

$$\tilde{Z}_h^t(s) - \dot{Z}_h^t(s) \leq 2b_h^t(s, a) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t(\pi^{t+1} G_{h+1}^t)(s, a) \leq G_h^t(s, a) = (\pi_h^{t+1} G_h^t)(s)$$

Finally, use optimism and the ring bridge: on $\mathcal{E}$, $Z_h^* \leq \tilde{Z}_h^t$ (lemma 6) and $\dot{Z}_h^t \leq Z_h^{\pi^{t+1}}$ (lemma 8), hence

$$Z_h^*(s) - Z_h^{\pi^{t+1}}(s) \leq \tilde{Z}_h^t(s) - \dot{Z}_h^t(s) \leq (\pi_h^{t+1} G_h^t)(s)$$

■

B.1.2. SAMPLING COMPLEXITY

Now we prove theorem 1 for $\beta > 0$

Proof the width certificate is:

$$G_h^t(s, a) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t \pi^{t+1} G_{h+1}^t(s) \right] \right\}$$

Let us transition to the true MDP. Using Bernstein inequality:

$$|(\tilde{p}_h^t - p_h) \pi^{t+1} G_{h+1}^t(s)| \leq \sqrt{2 \text{Var}_{p_h}(\pi^{t+1} G_{h+1}^t(s)) \alpha(n_h^t(s, a))} + \frac{2}{3} e^{\beta(H-h)} \alpha(n_h^t(s, a))$$

Now, we use the inequality $\text{Var}(\pi^{t+1} G_{h+1}^t(s)) \leq e^{\beta(H-h)} \pi^{t+1} G_{h+1}^t(s)$. Hence, using the inequality $\sqrt{xy} \leq x + y$:

$$|(\tilde{p}_h^t - p_h) \pi^{t+1} G_{h+1}^t(s)| \leq \frac{1}{H} p_h \pi^{t+1} G_{h+1}^t(s) + 3H e^{\beta(H-h)} \alpha(n_h^t(s, a))$$

And using the variance transportation lemmas 22,23 and using that $\alpha^*(n_h^t(s, a)) \leq \alpha(n_h^t(s, a))$:

$$\begin{aligned}
\sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} &\leq 2\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + \sqrt{4e^{\beta(H-h)} p_h(\tilde{Z}_{h+1}^t - Z_{h+1}^{\pi^{t+1}})(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \\
&\quad + 2e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\
&\leq 2\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4He^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\
&\quad + \frac{1}{H} p_h(\tilde{Z}_{h+1}^t - Z_{h+1}^{\pi^{t+1}})(s, a) + 2e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\
&\leq 2\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4He^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\
&\quad + \frac{1}{H} p_h \pi^{t+1} G_{h+1}^t(s) + 2e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}
\end{aligned}$$

Hence, by coarsening the constants for the sake of simplicity:

$$\begin{aligned}
b_h^t(s, a) &\leq 4\sqrt{2}\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4(2\sqrt{2} + 1)He^{\beta(H-h)} \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\
&\quad + (5 + 4\sqrt{2})e^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + \frac{2\sqrt{2}}{H} p_h \pi^{t+1} G_{h+1}^t(s) \\
&\leq 6\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + \frac{3}{H} p_h \pi^{t+1} G_{h+1}^t(s) + 27He^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}
\end{aligned}$$

We combine the two terms:

$$\begin{aligned}
G_h^t(s, a) &\leq e^{\beta r_h(s, a)} \left[12\sqrt{\text{Var}_{p_h}(Z_h^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + \frac{6}{H} p_h \pi^{t+1} G_{h+1}^t(s) \right. \\
&\quad + 54He^{\beta(H-h+1)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + \left(1 + \frac{2}{H}\right) p_h \pi^{t+1} G_{h+1}^t(s) \\
&\quad \left. + \left(1 + \frac{2}{H}\right) \frac{1}{H} p_h \pi^{t+1} G_{h+1}^t(s) + \left(1 + \frac{2}{H}\right) 3He^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \right]
\end{aligned}$$

Hence, simplifying it gives:

$$G_h^t(s, a) \leq e^{\beta r_h(s, a)} \left[12\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + \left(1 + \frac{11}{H}\right) p_h \pi^{t+1} G_{h+1}^t(s) + 63He^{\beta(H-h)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \right]$$

Unrolling this inequality and using the terminal condition $G_{H+1}^t = 0$ we get:

$$(\pi_1 G_1^t)(s_1) \leq \mathbb{E}^\pi \left[\sum_{h=1}^H \kappa^{h-1} \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(12 \sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})} \alpha^*(n_h^t(s_h, a_h) \wedge 1) \right. \right. \\ \left. \left. + 63H e^{\beta(H-h)} \alpha(n_h^t(s_h, a_h) \wedge 1) \right) \middle| s_1 \right]$$

where $k = 1 + \frac{7}{11}$. Since we have:

$$k^{h-1} = \left(1 + \frac{11}{H} \right)^{h-1} \leq \lim_{H \rightarrow +\infty} \left(1 + \frac{11}{H} \right)^H = e^{11}$$

we get:

$$(\pi_1 G_1^t)(s_1) \leq e^{11} \mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(12 \sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})} \alpha^*(n_h^t(s_h, a_h) \wedge 1) \right. \right. \\ \left. \left. + 63H e^{\beta(H-h)} \alpha(n_h^t(s_h, a_h) \wedge 1) \right) \middle| s_1 \right]$$

Now, we write the stopping rule:

$$(V_1^* - V_1^{\pi^{t+1}})(s_1) = \frac{1}{\beta} \log \left(\frac{Z_1^*}{Z_1^{\pi^{t+1}}} \right)(s_1) = \frac{1}{\beta} \log \left(1 + \frac{Z_1^* - Z_1^{\pi^{t+1}}}{Z_1^{\pi^{t+1}}} \right)(s_1) \leq \frac{1}{\beta} \frac{(Z_1^* - Z_1^{\pi^{t+1}})(s_1)}{Z_1^{\pi^{t+1}}(s_1)} \leq \frac{1}{\beta} \frac{\pi_1 G_1(s_1)}{Z_1^{\pi^{t+1}}}$$

Hence to have the policy be ε -optimal at time τ , we must have:

$$\pi_1^\tau G_1(s_1) \leq (e^{\beta\varepsilon} - 1) Z_1^{\pi^\tau}$$

But

$$\frac{1}{Z_1^{\pi^{t+1}}} (\pi_1 G_1^t)(s_1) \leq \\ e^{11} \mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(12 \sqrt{\frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \alpha(n_h^t(s_h, a_h) \wedge 1) \right. \right. \\ \left. \left. + 63H e^{\beta(H-h)} \alpha(n_h^t(s_h, a_h) \wedge 1) \right) \middle| s_1 \right]$$

Let us bound the first term, using Cauchy-Schwartz inequality:

$$\mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(\sqrt{\frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \left(\frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \right) \middle| s_1 \right] \\ = \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \left(\sqrt{\frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \left(\frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \right) \\ \leq \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp \left(2\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}}$$

For a policy π . By lemma ?? we have:

$$\sigma V_h^\pi(s) = e^{2\beta r_h(s, \pi(s))} \text{Var}_{p_h} (Z_{h+1}^\pi) (s, \pi(s)) + e^{2\beta r_h(s, \pi(s))} (p_h \sigma V_{h+1}^\pi) (s, \pi(s))$$

Multiplying the equation by $\sum_{i=1}^{h-1} r_i(s_i, a_i)$ and take the expectation under π :

$$\mathbb{E}^\pi \left[e^{2\beta \sum_{i=1}^{h-1} r_i(s_i, a_i)} \sigma V_h^\pi(s_h) \right] = \mathbb{E}^\pi \left[e^{2\beta \sum_{i=1}^h r_i(s_i, a_i)} \text{Var}_{p_h} (Z_{h+1}^\pi) \right] + \mathbb{E}^\pi \left[e^{2\beta \sum_{i=1}^h r_i(s_i, a_i)} \sigma V_{h+1}^\pi(s_{h+1}) \right]$$

By summing over $h = 1, \dots, H$ and since $\sigma V_{H+1}^\pi = 0$ we get:

$$\sum_{h=1}^H \mathbb{E}^\pi \left[e^{2\beta \sum_{i=1}^h r_i(s_i, a_i)} \frac{\text{Var}_{p_h} (Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2} \right] = \mathbb{E}^\pi \left[\frac{\sigma V_1^\pi(s_1)}{(Z_1^\pi)^2} \right]$$

But notice that for a deterministic policy π :

$$\frac{\sigma V_1^\pi(s_1)}{(Z_1^\pi)^2} = \frac{\text{Var}(L|s_1)}{\mathbb{E}(L|s_1)^2}$$

Where $L = \exp \left(\beta \sum_{i=1}^H r_i(s_i, a_i) \right)$. Using lemma 21 we get that:

$$\frac{\sigma V_1^\pi(s_1)}{(Z_1^\pi)^2} \leq e^{\beta H} - 1$$

Applying this for π^{t+1} we get:

$$\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp \left(2\beta \sum_{i=1}^h r_i(s_i, a_i) \right) \text{Var}_{p_h} (e^{\beta V_{h+1}^{\pi^{t+1}}}) \leq \frac{e^{\beta H} - 1}{4}$$

For the second term:

$$\begin{aligned} & \mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) H e^{\beta(H-h)} \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \right] \Big|_{s_1} \\ &= \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp \left(\beta \sum_{i=1}^h r_i(s_i, a_i) \right) H e^{\beta(H-h)} \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \\ &\leq \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) e^{\beta h} H e^{\beta(H-h)} \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \\ &\leq H e^{\beta H} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1 \right) \end{aligned}$$

Hence:

$$\begin{aligned}
 (\pi_1 G_1^t)(s_1) &\leq 12e^{11} \sqrt{(e^{\beta H} - 1) \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha^*(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)} \\
 &\quad + 63^7 e^{\beta H} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right) \\
 &\leq 12e^{11} \sqrt{(e^{\beta H} - 1)} \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha^*(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)} \\
 &\quad + 63^7 e^{\beta H} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)
 \end{aligned}$$

Let us sum over $t \leq \tau$. By sub-optimality for each episode $t = 0, \dots, \tau - 1$ we have:

$$\pi_1^\tau G_1(s_1) > (e^{\beta \varepsilon} - 1) Z_1^{\pi^\tau}$$

Hence by summing over all the episodes and using Cauchy-Schwartz:

$$\begin{aligned}
 \tau(e^{\beta \varepsilon} - 1) &\leq 12e^{11} \sum_{t=1}^{\tau-1} \sqrt{(e^{\beta H} - 1) \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha^*(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)} \\
 &\quad + 63e^{11} e^{\beta H} \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right) \\
 &\leq 12e^{11} \sqrt{(e^{\beta H} - 1)} \sqrt{\tau} \sqrt{\sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha^*(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)} \\
 &\quad + 63e^{11} e^{\beta H} \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)
 \end{aligned}$$

Using lemma 24 to relate the true counts to pseudo-counts we get:

$$\begin{aligned}
 \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right) &\leq \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \alpha(\tilde{n}_h^t(s,a) \vee 1) \\
 &\leq \alpha(\tau - 1, \delta) \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \frac{1}{\tilde{n}_h^t(s,a) \vee 1} \\
 &\leq \alpha(\tau - 1, \delta) \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} \frac{\tilde{n}_{h+1}^t(s,a) - \tilde{n}_h^t(s,a)}{\tilde{n}_h^t(s,a) \vee 1} \\
 &\leq 4HSA\alpha(\tau - 1, \delta) \log(\tau + 1)
 \end{aligned}$$

Where in the final inequality we used lemma 26. Similarly, we find:

$$\sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha^*(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right) \leq 4HSA\alpha^*(\tau - 1, \delta) \log(\tau + 1)$$

Therefore:

$$\tau(e^{\beta\varepsilon} - 1) \leq 12e^{11} \sqrt{(e^{\beta H} - 1)\tau HSA\alpha^*(\tau - 1, \delta) \log(\tau + 1) + 63e^{11}e^{\beta H} HSA\alpha(\tau - 1, \delta) \log(\tau + 1)}$$

Hence, by replacing α^* and α by their expression and using that $\log(\tau + 1) \leq \log(8e\tau)$ since $\tau \geq 1$:

$$\begin{aligned} \tau(e^{\beta\varepsilon} - 1) &\leq 12e^{11} \sqrt{(e^{\beta H} - 1)\tau HSA \left(\log\left(\frac{4SAH}{\delta}\right) \log(8e\tau) + \log(8e\tau)^2 \right)} \\ &\quad + 63e^{11}e^{\beta H} HSA \left(\log\left(\frac{4SAH}{\delta}\right) \log(8e\tau) + S \log(8e\tau)^2 \right) \end{aligned}$$

Finally, we use lemma 26 with :

$$\begin{aligned} C &= 12e^{11} \frac{\sqrt{(e^{|\beta|H} - 1)HSA}}{e^{\beta\varepsilon} - 1}, \quad A = \log\left(\frac{4SAH}{\delta}\right), \quad B = 1 \\ D &= \frac{63e^{11}e^{\beta H} H^2 SA}{e^{\beta\varepsilon} - 1} \quad \text{and} \quad E = S \end{aligned}$$

Which yield:

$$\tau \leq \frac{(e^{|\beta|H} - 1)HSA}{(e^{\beta\varepsilon} - 1)^2} \left(\log\left(\frac{4SAH}{\delta}\right) + 1 \right) C_1^2 + 3 \frac{e^{\beta H} H^2 SA}{e^{\beta\varepsilon} - 1} \left(\log\left(\frac{4SAH}{\delta}\right) + S \right) C_1^2 + 1$$

Where $C_1 = \frac{8}{5} \log \left(4e^{17} \frac{(S+1)(H+1)e^{|\beta H} SAH^2}{e^{\beta\varepsilon} - 1} \right)$

In particular, if $\varepsilon \leq \frac{1}{\beta} \log(1 + \frac{1}{HS}) \leq \frac{1}{|\beta|HS}$ then:

$$\tau \leq \frac{(e^{|\beta|H} - 1)HSA}{(e^{\beta\varepsilon} - 1)^2} \log\left(\frac{4SAH}{\delta}\right) C_2^2$$

Where $C_2 = 3C_1$. We can finally hide the constants and the log terms to get:

$$\tau = \tilde{O} \left(\frac{(e^{|\beta|H} - 1)HSA}{(e^{|\beta|\varepsilon} - 1)^2} \right)$$

■

B.2. Case $\beta < 0$

B.2.1. CONFIDENCE BOUNDS

We first start with a lemma:

Lemma 10 *On the good event $\mathcal{E}^-$:*

$$\begin{aligned} |(p_h - \tilde{p}_h^t) Z_{h+1}^*| &\leq 2\sqrt{2} \sqrt{\text{Var}_{\tilde{p}_h^t}(\mathcal{Z}_h^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 5(1 - e^{\beta(H-h)}) \alpha(n_h^t(s, a)) \\ &\quad + 4H(1 - e^{\beta(H-hH)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} + \frac{1}{H} \tilde{p}_h^t \left(Z_{h+1}^* - \mathcal{Z}_{h+1}^t \right)(s, a) \end{aligned}$$

Proof On the good event $\mathcal{E}^-$ we have:

$$|(p_h - \tilde{p}_h^t)Z_{h+1}^*(s, a)| \leq \sqrt{2 \text{Var}_{p_h}(Z_{h+1}^*) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 3(1 - e^{\beta(H-h)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}$$

Since Z_h^* and p_h are unknown, we use a variance transportation inequality to replace with its value for $\tilde{Z}_h^t$ the optimistic bound for V^* . By applying lemma 22 and lemma 23 successively:

$$\begin{aligned} \text{Var}_{p_h}(Z_{h+1}^*) &\leq 2 \text{Var}_{\tilde{p}_h^t}(Z_{h+1}^*)(s, a) + 4(1 - e^{\beta(H-h)})^2 \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\ &\leq 4 \text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t)(s, a) + 4(1 - e^{\beta(H-h)}) \tilde{p}_h^t(Z_{h+1}^* - \tilde{Z}_{h+1}^t)(s, a) + 4(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \end{aligned}$$

Hence, using the inequality $\sqrt{a+b} \leq \sqrt{a} + \sqrt{b}$ and then $\sqrt{ab} \leq a\eta + \frac{b}{\eta}$ for $\eta = H$ and using that $\alpha^*(n_h^t(s, a)) \leq \alpha(n_h^t(s, a))$:

$$\begin{aligned} \sqrt{\text{Var}_{p_h}(Z_h^*)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} &\leq 2\sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_h^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \\ &\quad + \sqrt{4(1 - e^{\beta(H-h)}) \tilde{p}_h^t(Z_{h+1}^* - \tilde{Z}_{h+1}^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \\ &\quad + 2(1 - e^{\beta(H-h-1)}) \sqrt{\frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} \\ &\leq 2\sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_h^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4H(1 - e^{\beta(H-h)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\ &\quad + \frac{1}{H} \tilde{p}_h^t(Z_{h+1}^* - \tilde{Z}_{h+1}^t)(s, a) + 2(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \end{aligned}$$

Hence:

$$\begin{aligned} |p_h Z_{h+1}^* - \tilde{p}_h^t(s, a) Z_{h+1}^*| &\leq 2\sqrt{2} \sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_{h+1}^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 5(1 - e^{\beta(H-h)}) \alpha(n_h^t(s, a)) \\ &\quad + 4H(1 - e^{\beta(H-h)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} + \frac{1}{H} \tilde{p}_h^t(Z_{h+1}^* - \tilde{Z}_{h+1}^t)(s, a) \end{aligned}$$

■

Denote the bonus term as:

$$b_h^t(s, a) = 2\sqrt{2} \sqrt{\text{Var}_{\tilde{p}_h^t}(\tilde{Z}_h^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 5(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + 4H(1 - e^{\beta(H-h)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}$$

Now, like the case $\beta > 0$ we define optimistic and pessimistic state-value function on the exponential transform of Q_h^* which denoted by U_h^* .

$$\tilde{U}_h^t(s, a) = \min \left\{ 1, e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - Z_{h+1}^t)(s, a) \right] \right\}$$

And

$$\tilde{Z}_h^t(s) = \min_{a \in \mathcal{A}} \tilde{U}_h^t(s, a)$$

$$\tilde{Z}_{H+1}^t(s) = 1$$

And a pessimistic lower bound

$$\underline{U}_h^t(s, a) = \max \left\{ e^{\beta(H-h-1)}, e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \underline{Z}_{h+1}^t(s, a) - b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \underline{Z}_{h+1}^t)(s, a) \right] \right\}$$

And

$$\underline{Z}_h^t(s) = \min_{a \in \mathcal{A}} \underline{U}_h^t(s, a)$$

$$\underline{Z}_{H+1}^t(s) = 1$$

We consider the greedy policy:

$$\pi_h^{t+1}(s) = \arg \min_{a \in \mathcal{A}} \underline{U}_h^t(s, a)$$

Let us prove an optimism lemma:

Lemma 11 *On the good event $\mathcal{E}^-$ we have:*

$$\underline{U}_h^t(s, a) \leq U_h^*(s, a) \leq \tilde{U}_h^t(s, a)$$

and

$$\underline{Z}_h^t(s) \leq Z_h^*(s) \leq \tilde{Z}_h^t(s)$$

Proof We proceed by induction over h . For $h = H + 1$ the result is trivially upper bounding and (resp. lower bounding) Q^* by H and 1.

Assume the inequality holds for $h' > h$. Fix (s, a) and assume $\tilde{Q}_h^t(s, a) < H$. we have that:

$$\begin{aligned} \tilde{U}_h^t(s, a) - U_h^*(s, a) &= e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \underline{Z}_{h+1}^t)(s, a) - p_h Z_{h+1}^*(s, a) \right] \\ &= e^{\beta r_h(s, a)} \left[\tilde{p}_h^t (\tilde{Z}_{h+1}^t(s, a) - Z_{h+1}^*(s, a)) + (\tilde{p}_h^t - p_h) Z_{h+1}^*(s, a) \right. \\ &\quad \left. + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \underline{Z}_{h+1}^t)(s, a) \right] \end{aligned}$$

But we know by Bernstein inequality that:

$$\left(\tilde{p}_h^t - p_h\right) Z_h^*(s, a) \geq -b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t \left(Z_{h+1}^* - \tilde{Z}_{h+1}^t\right)(s, a) \geq -b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - Z_{h+1}^*\right)(s, a)$$

Hence:

$$\tilde{U}_h^t(s, a) - U_h^*(s, a) \geq e^{\beta r_h(s, a)} \left[\left(1 + \frac{1}{H}\right) \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t(s, a) - Z_{h+1}^*(s, a)\right) \right] \geq 0$$

Where we used the induction hypothesis. We prove the pessimistic property in the same way $\blacksquare$

B.2.2. STOPPING RULE

We define the stopping rule for the algorithm for $\beta < 0$ as:

$$G_h^t(s, a) = \min \left\{ 1, e^{\beta r_h(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t \pi^{t+1} G_{h+1}^t(s, a) \right] \right\}$$

Lemma 12 establishes the validity of this stopping rule by showing that, with high probability, it bounds the certificate width:

Lemma 12 *On the good event $\mathcal{E}^+$, for all t and all h ,*

$$Z_h^*(s) - Z_h^{\pi^{t+1}}(s) \leq \pi_h^{\pi^{t+1}} G_h^t(s) \quad \forall s \in \mathcal{S}$$

In particular, at the initial state s_1 , $V_1^(s_1) - V_1^{\pi^{t+1}}(s_1) \leq \pi_1^{\pi^{t+1}} G_1^t(s_1)$*

We prove the lemma 12 in this section :

Like the case $\beta > 0$ we introduce the analysis-tool $\tilde{U}$. Due to β being negative, now we cannot compare $\tilde{Z}_h^t$ and $Z_h^{\pi^{t+1}}$. As in the case $\beta > 0$, we define the exponential ring quantities $\tilde{U}_h^t$ and $\tilde{Z}_h^t$. They are define as before to satisfy the exponential Bellman equation while at the same time upper bounding both the $\tilde{U}_h^t$ and $U_h^{\pi^{t+1}}$:

- Terminal: $\tilde{Z}_{H+1}^t(s) = 1$ for all s .
- For $h = H, H-1, \dots, 1$:

1. Define the “true-backup candidate”

$$\tilde{U}_{h, \text{true}}^t(s, a) := e^{\beta r_h(s, a)} (p_h \tilde{Z}_{h+1}^t)(s, a)$$

2. Define a “optimistic-backup candidate” $\tilde{U}_{h, \text{opt}}^t(s, a)$ by taking **the same functional form as the pessimistic estimator**, but plugging $\tilde{Z}_{h+1}^t$ wherever the estimator uses a next-step value input:

$$\tilde{U}_{h, \text{opt}}^t(s, a) := \min \left\{ 1, e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \tilde{Z}_{h+1}^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \tilde{Z}_{h+1}^t \right)(s, a) \right] \right\}$$

3. Then set

$$\mathring{U}_h^t(s, a) := \max \left\{ \mathring{U}_{h, \text{true}}^t(s, a), \mathring{U}_{h, \text{opt}}^t(s, a) \right\}$$

4. Finally define the ring value **along the greedy policy**:

$$\mathring{Z}_h^t(s) := \mathring{U}_h^t(s, \pi_h^{t+1}(s))$$

Lemma 13 For all t , For all (h, s, a) :

$$\mathring{U}_h^t(s, a) \geq \max \left(\tilde{U}_h^t(s, a), U_h^{\pi_h^{t+1}}(s, a) \right)$$

and

$$\mathring{Z}_h^t(s) \geq \max \left(\tilde{Z}_h^t(s), Z_h^{\pi_h^{t+1}}(s, a) \right)$$

Proof We proceed by backward induction. For $h = H + 1$, all values are equal to 0 so the inequalities hold. Assume that for some $h \leq H$ we have for all (s, a) :

$$\mathring{U}_{h+1}^t(s, a) \geq \max \left(\tilde{U}_{h+1}^t(s, a), U_{h+1}^{\pi_{h+1}^{t+1}}(s, a) \right)$$

and

$$\mathring{Z}_{h+1}^t(s) \geq \max \left(\tilde{Z}_{h+1}^t(s), Z_{h+1}^{\pi_{h+1}^{t+1}}(s, a) \right)$$

we have by construction:

$$\mathring{U}_h^t(s, a) \geq \mathring{U}_{h, \text{true}}^t(s, a) = e^{\beta r_h(s, a)} (p_h \mathring{Z}_{h+1}^t)(s, a) \geq e^{\beta r_h(s, a)} (p_h Z_{h+1}^{\pi_{h+1}^{t+1}})(s, a) = U_h^{\pi_h^{t+1}}(s, a)$$

Where we used the induction hypothesis and the monotonicity of the exponential Bellman operator.

$$\begin{aligned} \mathring{U}_h^t(s, a) - \tilde{U}_h^t(s, a) &\geq \mathring{U}_{h, \text{opt}}^t(s, a) - \tilde{U}_h^t(s, a) \\ &\geq e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \mathring{Z}_{h+1}^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\mathring{Z}_{h+1}^t - \mathring{Z}_{h+1}^t)(s, a) \right. \\ &\quad \left. - \left(\tilde{p}_h^t \tilde{Z}_h^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathring{Z}_{h+1}^t)(s, a) \right) \right] \\ &\geq e^{\beta r_h(s, a)} \left[\left(1 + \frac{1}{H} \right) \tilde{p}_h^t (\mathring{Z}_{h+1}^t - \tilde{Z}_{h+1}^t)(s, a) \right] \end{aligned}$$

Where we applied the induction hypothesis, we conclude then:

$$\mathring{U}_h^t(s, a) \geq \tilde{U}_h^t(s, a)$$

For Z :

$$\mathring{Z}_h^t(s) - Z_h^{\pi_h^{t+1}}(s, a) = \mathring{U}_h^t(s, \pi_h^t(s)) - U_h^{\pi_h^{t+1}}(s, \pi_h^{t+1}(s)) \geq 0$$

And:

$$\mathring{Z}_h^t(s) = \mathring{U}_h^t(s, \pi_h^t(s)) \geq \tilde{U}_h^t(s, \pi_h^t(s)) \geq \min_{a \in \mathcal{A}} \tilde{U}_h^t(s, a) = \tilde{Z}_h^t(s, a)$$

Which conclude the recurrence ■

Lemma 14 For any t , any $h \in [H]$ and any state-action pair:

$$\dot{U}_h^t(s, a) - \underline{U}_h^t(s, a) \leq e^{\beta r_h(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t \left(\dot{Z}_h^t(s, a) - \underline{Z}_{h+1}^t(s, a) \right) \right]$$

Proof Fix $h \in [H]$ and fix a state-action pair (s, a) . We have two cases:

First case: $\dot{U}_h^t(s, a) = \dot{U}_{h, \text{true}}^t(s, a)$, we have:

$$\dot{U}_h^t(s, a) - \underline{U}_h^t(s, a) \leq e^{\beta r_h(s, a)} \left[b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t \left(\tilde{Z}_{h+1}^t - \underline{Z}_{h+1}^t \right)(s, a) + p_h \dot{Z}_{h+1}^t(s, a) - \tilde{p}_h^t \underline{Z}_{h+1}^t(s, a) \right]$$

The last term can be written as:

$$p_h \dot{Z}_{h+1}^t(s, a) - \tilde{p}_h^t \underline{Z}_{h+1}^t(s, a) = \tilde{p}_h^t \left(\dot{Z}_{h+1}^t(s, a) - \underline{Z}_{h+1}^t(s, a) \right) + (p_h - \tilde{p}_h^t) \dot{Z}_h^t(s, a)$$

For the second term, on the event (F_V^c) we have:

$$|(p_h - \tilde{p}_h^t) \dot{Z}_{h+1}^t(s, a)| \leq \sqrt{2 \text{Var}(\dot{Z}_{h+1}^t)(s, a) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} + 3(1 - e^{\beta(H-h-1)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}$$

We use the variance transportation lemmas 22 and 23 and the inequality $\sqrt{xy} \leq x + y$ as before:

$$\begin{aligned} \sqrt{\text{Var}(\dot{Z}_{h+1}^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} &\leq 2\sqrt{\text{Var}(\tilde{Z}_{h+1}^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4H(1 - e^{\beta(H-h-1)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\ &\quad + 5(1 - e^{\beta(H-h-1)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} + \frac{1}{H} \tilde{p}_h^t (\dot{Z}_{h+1}^t(s, a) - \tilde{Z}_{h+1}^t)(s, a) \\ &\leq 2\sqrt{\text{Var}(\tilde{Z}_{h+1}^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4H(1 - e^{\beta(H-h-1)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\ &\quad + 5(1 - e^{\beta(H-h-1)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} + \frac{1}{H} \tilde{p}_h^t (\dot{Z}_{h+1}^t(s, a) - \underline{Z}_{h+1}^t)(s, a) \end{aligned}$$

Hence:

$$|(p_h - \tilde{p}_h^t) \dot{Z}_{h+1}^t(s, a)| \leq b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\dot{Z}_h^t(s, a) - \underline{Z}_{h+1}^t)(s, a)$$

Hence we get:

$$p_h \dot{Z}_{h+1}^t(s, a) - \tilde{p}_h^t \underline{Z}_{h+1}^t(s, a) \leq b_h^t(s, a) + \left(1 + \frac{1}{H}\right) \tilde{p}_h^t \left(\dot{Z}_h^t(s, a) - \underline{Z}_{h+1}^t(s, a) \right)$$

Hence by substituting and using lemma 13, we have:

$$\begin{aligned} \dot{U}_h^t(s, a) - \underline{U}_h^t(s, a) &\leq e^{\beta r_h(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t \left(\dot{Z}_h^t(s, a) - \underline{Z}_{h+1}^t(s, a) \right) \right] \\ &\leq e^{\beta r_h(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H}\right) \tilde{p}_h^t \left(\dot{Z}_h^t(s, a) - \underline{Z}_{h+1}^t(s, a) \right) \right] \end{aligned}$$

Second case: $\mathring{U}_h^t(s, a) = \mathring{U}_{h,\text{opt}}^t(s, a)$, we have:

$$\begin{aligned} \mathring{U}_h^t(s, a) - \mathcal{U}_h^t(s, a) &\leq e^{\beta r_h(s, a)} \left[\tilde{p}_h^t \mathring{Z}_{h+1}^t(s, a) + b_h^t(s, a) + \frac{1}{H} \tilde{p}_h^t (\mathring{Z}_{h+1}^t - \mathcal{Z}_{h+1}^t)(s, a) \right. \\ &\quad \left. - \left(\tilde{p}_h^t \mathcal{Z}_{h+1}^t(s, a) - b_h^t(s, a) - \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathcal{Z}_{h+1}^t)(s, a) \right) \right] \\ &= e^{\beta r_h(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{1}{H} \right) \tilde{p}_h^t (\mathring{Z}_{h+1}^t - \mathcal{Z}_{h+1}^t)(s, a) + \frac{1}{H} \tilde{p}_h^t (\tilde{Z}_{h+1}^t - \mathcal{Z}_{h+1}^t)(s, a) \right] \end{aligned}$$

Using lemma 13 we get:

$$\mathring{U}_h^t(s, a) - \mathcal{U}_h^t(s, a) \leq e^{\beta r_h(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \tilde{p}_h^t (\mathring{Z}_h^t(s, a) - \mathcal{Z}_{h+1}^t)(s, a) \right]$$

■

We now prove lemma 12:

Proof We first prove by backward induction that, for all h and s ,

$$\mathring{Z}_h^t(s) - \mathcal{Z}_h^t(s) \leq (\pi_h^{t+1} G_h^t)(s).$$

For $h = H + 1$ it holds since both sides are 0. Assume it holds at step $h + 1$. For $a = \pi_h^{t+1}(s)$

$$\mathring{Z}_h^t(s) - \mathcal{Z}_h^t(s) = \mathring{U}_h^t(s, a) - \mathcal{U}_h^t(s, a)$$

Apply Lemma 13:

$$\mathring{U}_h^t(s, a) - \mathcal{U}_h^t(s, a) \leq e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \tilde{p}_h^t (\mathring{Z}_h^t(s, a) - \mathcal{Z}_{h+1}^t)(s, a) \right]$$

By the induction hypothesis:

$$\tilde{p}_h^t (\mathring{Z}_h^t(s, a) - \mathcal{Z}_{h+1}^t)(s, a) \leq \tilde{p}_h^t (\pi_h^{t+1} G_{h+1}^t)(s, a).$$

Thus,

$$\mathring{Z}_h^t(s) - \mathcal{Z}_h^t(s) \leq 2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \tilde{p}_h^t (\pi_h^{t+1} G_{h+1}^t)(s, a) \leq G_h^t(s, a) = (\pi_h^{t+1} G_h^t)(s)$$

Finally, use optimism and the ring bridge: on $\mathcal{E}$, $Z_h^\star \geq \mathcal{Z}_h^t$ (optimism lemma 11) and $Z_h^{\pi^{t+1}} \leq \mathring{Z}_h^t$ (Lemma 13):

$$Z_h^{\pi^{t+1}}(s) - Z_h^\star(s) \leq \mathring{Z}_h^t(s) - \mathcal{Z}_h^t(s) \leq (\pi_h^{t+1} G_h^t)(s)$$

■

B.2.3. SAMPLING COMPLEXITY

Proof [Theorem 1] the width certificate is:

$$G_h^t(s, a) = \min \left\{ e^{\beta(H-h-1)}, e^{\beta r_h^t(s, a)} \left[2b_h^t(s, a) + \left(1 + \frac{2}{H} \right) \hat{p}_h^t \pi^{t+1} G_{h+1}^t(s) \right] \right\}$$

Let us transition to the true MDP. Using Bernstein inequality:

$$|(\hat{p}_h^t - p_h) \pi^{t+1} G_{h+1}^t(s)| \leq \sqrt{2 \text{Var}_{p_h}(\pi^{t+1} G_{h+1}^t(s)) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} + \frac{2}{3} (1 - e^{\beta(H-h-1)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}$$

Now, we use the inequality $\text{Var}(\pi_{h+1}^{t+1} G_{h+1}^t(s)) \leq (1 - e^{\beta(H-h)}) \pi_{h+1}^{t+1} G_{h+1}^t(s)$. Hence, using the inequality $\sqrt{xy} \leq x + y$:

$$|(\hat{p}_h^t - p_h) \pi^{t+1} G_{h+1}^t(s)| \leq \frac{1}{H} p_h \pi^{t+1} G_{h+1}^t(s) + 3H(1 - e^{\beta(H-h+1)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}$$

And using the variance transportation lemmas 22 and 23 and that $\alpha^*(n_h^t(s, a)) \leq \alpha(n_h^t(s, a))$:

$$\begin{aligned} \sqrt{\text{Var}_{\hat{p}_h^t}(\tilde{Z}_{h+1}^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} &\leq 2\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \\ &\quad + \sqrt{4(1 - e^{\beta(H-h)}) p_h (Z_{h+1}^{\pi^{t+1}} - \tilde{Z}_{h+1}^t)(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \\ &\quad + 2(1 - e^{\beta(H-h-1)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\ &\leq 2\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4H(1 - e^{\beta(H-h)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\ &\quad + \frac{1}{H} p_h (Z_{h+1}^{\pi^{t+1}} - \tilde{Z}_{h+1}^t)(s, a) + 2(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \\ &\leq 2\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + 4H(1 - e^{\beta(H-h)}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)} \\ &\quad + \frac{1}{H} p_h \pi^{t+1} G_{h+1}^t(s) + 2(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \end{aligned}$$

Since we have :

$$Z_{h+1}^{\pi^{t+1}} - \tilde{Z}_{h+1}^t \leq \hat{Z}_{h+1}^t - \tilde{Z}_{h+1}^t \leq G_{h+1}^t(s)$$

Hence, using that $\alpha^*(n_h^t(s, a)) \leq \alpha(n_h^t(s, a))$ and simplifying constants and using that $H \geq 1$:

$$b_h^t(s, a) \leq 6\sqrt{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + \frac{3}{H} p_h \pi^{t+1} G_{h+1}^t(s) + 27H(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}$$

We combine the two terms:

$$\begin{aligned}
G_h^t(s, a) \leq e^{\beta r_h(s, a)} & \left[12 \sqrt{\text{Var}_{p_h}(Z_h^{\pi^{t+1}}) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} + \frac{6}{H} p_h \pi^{t+1} G_{h+1}^t(s) \right. \\
& + 54H(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} + \left(1 + \frac{2}{H}\right) p_h \pi^{t+1} G_{h+1}^t(s) \\
& \left. + \left(1 + \frac{2}{H}\right) \frac{1}{H} p_h \pi^{t+1} G_{h+1}^t(s) + \left(1 + \frac{2}{H}\right) 3H(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \right]
\end{aligned}$$

Hence, simplifying it gives:

$$G_h^t(s, a) \leq e^{\beta r_h(s, a)} \left[12 \sqrt{\text{Var}_{p_h}(Z_h^{\pi^{t+1}}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} + \left(1 + \frac{11}{H}\right) p_h \pi^{t+1} G_{h+1}^t(s) + 63H(1 - e^{\beta(H-h)}) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \right]$$

Unrolling this inequality like the case $\beta > 0$:

$$\begin{aligned}
(\pi_1 G_1^t)(s_1) \leq e^{11} \mathbb{E}^\pi & \left[\sum_{h=1}^H \exp\left(\beta \sum_{i=1}^h r_i(s_i, a_i)\right) \left(12 \sqrt{\text{Var}_{p_h}(Z_{h+1}^\pi) \alpha(n_h^t(s_h, a_h) \wedge 1)} \right. \right. \\
& \left. \left. + 63H(1 - e^{\beta(H-h)}) \alpha(n_h^t(s_h, a_h) \wedge 1) \right) \right] \Big| s_1
\end{aligned}$$

Now, we write:

$$(V_1^\star - V_1^{\pi^{t+1}})(s_1) = \frac{1}{\beta} \log\left(\frac{Z_1^\star}{Z_1^{\pi^{t+1}}}\right)(s_1) = \frac{1}{\beta} \log\left(1 + \frac{Z_1^\star - Z_1^{\pi^{t+1}}}{Z_1^{\pi^{t+1}}}\right)(s_1)$$

But since $Z_1^{\pi^{t+1}} - Z_1^\star \leq \pi_1 G_1(s)$ then:

$$(V_1^\star - V_1^{\pi^{t+1}})(s_1) \leq \frac{1}{\beta} \log\left(1 - \frac{\pi_1 G_1(s)}{Z_1^{\pi^{t+1}}(s)}\right)$$

Hence to have the policy be ε -optimal, we must have:

$$\pi_1^\tau G_1(s_1) \leq (1 - e^{\beta\varepsilon}) Z_1^{\pi^\tau}$$

$$\begin{aligned}
\frac{1}{Z_1^{\pi^{t+1}}} (\pi_1 G_1^t)(s_1) \leq \frac{e^{11}}{\beta} \mathbb{E}^{\pi^{t+1}} & \left[\sum_{h=1}^H \exp\left(\beta \sum_{i=1}^h r_i(s_i, a_i)\right) \left(12 \sqrt{\frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2} \left(\frac{\alpha(n_h^t(s_h, a_h))}{n_h^t(s, a)} \wedge 1\right)} \right. \right. \\
& \left. \left. + 63H(1 - e^{\beta(H-h)}) \left(\frac{\alpha(n_h^t(s_h, a_h))}{n_h^t(s, a)} \wedge 1\right) \right) \right] \Big| s_1
\end{aligned}$$

Like the case $\beta > 0$ we write directly:

$$\sum_{h=1}^H \sum_{s, a} p_h^{t+1}(s, a) \exp\left(2\beta \sum_{i=1}^h r_i(s_i, a_i)\right) \frac{\text{Var}_{p_h}(Z_{h+1}^\pi)}{(Z_1^{\pi^{t+1}})^2} = \mathbb{E}^\pi \left[\frac{\sigma V_1^{\pi^{t+1}}}{(Z_1^{\pi^{t+1}})^2} \right]$$

But since the greedy policy is deterministic we have:

$$\frac{\sigma V_1^\pi(s_1)}{(Z_1^\pi)^2} = \frac{\text{Var}(L|s_1)}{\mathbb{E}(L|s_1)^2}$$

Where $L = e^{\beta \sum_{i=1}^H r_i(s_i, a_i)}$, using lemma 21 we get that:

$$\frac{\sigma V_1^\pi(s_1)}{(Z_1^\pi)^2} \leq e^{|\beta|H} - 1$$

Hence:

$$\begin{aligned} & \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp\left(\beta \sum_{i=1}^h r_i(s_i, a_i)\right) \left(12 \sqrt{\frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1\right)\right) \\ & \leq 12 \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp\left(2\beta \sum_{i=1}^h r_i(s_i, a_i)\right) \frac{\text{Var}_{p_h}(Z_{h+1}^{\pi^{t+1}})}{(Z_1^{\pi^{t+1}})^2}} \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)}} \\ & \leq 12 \sqrt{e^{|\beta|H} - 1} \sqrt{\sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \frac{\alpha^*(n_h^t(s, a))}{n_h^t(s, a)}} \end{aligned}$$

For the second term:

$$\begin{aligned} & \frac{1}{Z_1^{\pi^{t+1}}(s_1)} \mathbb{E}^{\pi^{t+1}} \left[\sum_{h=1}^H \exp\left(\beta \sum_{i=1}^h r_i(s_i, a_i)\right) H(1 - e^{\beta(H-h)}) \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1\right) \right] \Big| s_1 \\ & = \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp\left(\beta \sum_{i=1}^h r_i(s_i, a_i)\right) H(1 - e^{\beta(H-h)}) \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1\right) \\ & \leq \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \exp\left(\beta(h-1)\right) H(1 - e^{\beta(H-h)}) \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1\right) \\ & \leq H e^{\beta H} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s, a) \left(\frac{\alpha(n_h^t(s, a))}{n_h^t(s, a)} \wedge 1\right) \end{aligned}$$

We then sum on $t < \tau$ and use that by sub-optimality we have for $t = 1, \dots, \tau - 1$:

$$\frac{\pi_1^{t+1} G_1(s)}{Z_1^{\pi^{t+1}}} \geq 1 - e^{\beta \varepsilon}$$

$$\begin{aligned}
\tau(1 - e^{\beta\varepsilon}) &\leq 12e^{11} \sum_{t=1}^{\tau-1} \sqrt{(e^{|\beta|H} - 1) \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha^*(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)} \\
&\quad + 63e^{11} H e^{|\beta|H} \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right) \\
&\leq 12e^{11} \sqrt{(e^{|\beta|H} - 1)} \sqrt{T} \sqrt{\sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha^*(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)} \\
&\quad + 63e^{11} H e^{|\beta|H} \sum_{t=1}^{\tau-1} \sum_{h=1}^H \sum_{s,a} p_h^{t+1}(s,a) \left(\frac{\alpha(n_h^t(s,a))}{n_h^t(s,a)} \wedge 1 \right)
\end{aligned}$$

Similarly to the case $\beta > 0$ we bound the other terms using the counting argument which yield:

$$\tau(1 - e^{\beta\varepsilon}) \leq 12e^{11} \sqrt{(e^{|\beta|H} - 1) \tau H S A \alpha^*(\tau - 1, \delta) \log(\tau + 1)} + 63e^{11} e^{|\beta|H} H^2 S A \alpha(\tau - 1, \delta) \log(\tau + 1)$$

We replace α^* and α by their expressions and using that $\log(\tau + 1) \leq \log(8e\tau)$ since $\tau \geq 1$:

$$\begin{aligned}
\tau(1 - e^{\beta\varepsilon}) &\leq 12e^{11} \sqrt{(e^{|\beta|H} - 1) \tau H S A \left(\log\left(\frac{4SAH}{\delta}\right) \log(8e\tau) + \log(8e\tau)^2 \right)} \\
&\quad + 63e^{11} e^{|\beta|H} H^2 S A \left(\log\left(\frac{4SAH}{\delta}\right) \log(8e\tau) + S \log(8e\tau)^2 \right)
\end{aligned}$$

Finally, we use lemma 26 with :

$$\begin{aligned}
C &= 12e^{11} \frac{\sqrt{(e^{|\beta|H} - 1) H S A}}{1 - e^{\beta\varepsilon}}, \quad A = \log\left(\frac{4SAH}{\delta}\right), \quad B = 1 \\
D &= \frac{63e^{11} e^{\beta H} H^2 S A}{1 - e^{\beta\varepsilon}} \quad \text{and} \quad E = S
\end{aligned}$$

Which yield:

$$\tau \leq \frac{(e^{|\beta|H} - 1) H S A}{(1 - e^{\beta\varepsilon})^2} \left(\log\left(\frac{4SAH}{\delta}\right) + 1 \right) C_1^2 + 3 \frac{e^{\beta H} H^2 S A}{1 - e^{\beta\varepsilon}} \left(\log\left(\frac{4SAH}{\delta}\right) + S \right) C_1^2 + 1$$

Where $C_1 = \frac{8}{5} \log \left(4e^{17} \frac{(S+1)(H+1)e^{|\beta H} S A H^2}{1 - e^{\beta\varepsilon}} \right)$

In particular, assuming that $HS \geq 2$: if $\varepsilon \leq \frac{1}{|\beta|} \log\left(\frac{HS}{HS-1}\right) \leq \frac{2}{|\beta|HS}$ then:

$$\tau \leq \frac{(e^{|\beta|H} - 1) H S A}{(1 - e^{\beta\varepsilon})^2} \log\left(\frac{4SAH}{\delta}\right) C_2^2$$

Where $C_2 = 3C_1$. We can finally hide the constants and the log terms to get:

$$\tau = \tilde{O} \left(\frac{(e^{|\beta|H} - 1) H S A}{(1 - e^{\beta\varepsilon})^2} \right) = \tilde{O} \left(\frac{(e^{|\beta|H} - 1) H S A}{(e^{|\beta|\varepsilon} - 1)^2} \right)$$

Since $\frac{1}{1 - e^{\beta\varepsilon}} = \frac{e^{|\beta|\varepsilon}}{e^{|\beta|\varepsilon} - 1} \leq \frac{2}{e^{|\beta|\varepsilon} - 1}$ ■

Appendix C. Lower bound

We first state a change of measure for bandit models result from (Kaufmann et al., 2016):

Lemma 15 *Let $N_a(t) = \sum_{s=1}^t \mathbb{1}_{\{A_s=a\}}$ be the number of draws of arm a between the instants 1 and t and $N_a = N_a(\tau)$ be the total number of draws of arm a by some algorithm $\mathcal{A} = ((A_t), \tau, \hat{S}_m)$.*

Let ν and ν' be two bandit models with K arms such that for all a , the distributions ν_a and ν'_a are mutually absolutely continuous. For any almost-surely finite stopping time σ with respect to $(\mathcal{F}_t)$,

$$\sum_{a=1}^K \mathbb{E}_\nu[N_a(\sigma)] \text{KL}(\nu_a, \nu'_a) \geq \sup_{\mathcal{E} \in \mathcal{F}_\sigma} d(\mathbb{P}_\nu(\mathcal{E}), \mathbb{P}_{\nu'}(\mathcal{E}))$$

where $d(x, y) := x \log(x/y) + (1-x) \log((1-x)/(1-y))$ is the binary relative entropy, with the convention that $d(0, 0) = d(1, 1) = 0$.

We then prove theorem :

Proof We introduce the following MDP: We start at state s_0 . We pick a random number uniformly from $\{1, \dots, n\}$ where $n = S - 3$ and we jump to bandit i . The learned take an action $a \in \{1, \dots, A\}$ to jump either to $+$ to receive reward 1 or jump to $-$ and receive reward 0 and stays there for the rest of the episode. That means that if we take an optimal choice we receive a reward $H - 2$ at the end, otherwise we receive 0.

For each bandit i there is one optimal arm a_i^* that takes us to $+$ with probability p_+ and $A - 1$ arms that takes us to $-$ with probability p_- . Which arm is optimal for i define a vector $I \in [A]^n$. We constructed a family of MDPs and each MDP is identifiable by the vector I i.e which arms are optimal for each bandit.

The bandit-level entropic value function is :

$$V(p) = \frac{1}{\beta} \log(\mathbb{E}[e^{\beta R}]) = \frac{1}{\beta} \log((1-p) \cdot 1 + p \cdot e^{\beta(H-2)}) = \frac{1}{\beta} \log(1 + cp)$$

Where $c = e^{\beta(H-2)} - 1$. We define the transition probabilities as :

$$p_- = \frac{1}{2c+1} \quad p_+ = p_-(1+\eta)$$

We choose η so that the per-bandit entropic gap is at very small so that it is hard to identify the optimal action, we get :

$$V(p_+) - V(p_-) = \frac{1}{\beta} \log \frac{1 + cp_+}{1 + cp_-}$$

Let $x := cp_- = \frac{c}{2(c+1)}$ Then

$$\frac{1 + cp_+}{1 + cp_-} = \frac{1 + x(1+\eta)}{1 + x} = 1 + \frac{x}{1+x} \eta$$

A short algebra gives

$$\frac{x}{1+x} = \frac{c}{3c+2}$$

Therefore

$$V(p_+) - V(p_-) = \frac{1}{\beta} \log \left(1 + \frac{c}{3c+2} \eta \right)$$

Fix a target gap $\varepsilon_{\text{step}} = \kappa \varepsilon$, $\varepsilon_{\text{step}}$ is how much we can afford to be wrong in each bandit. Choose η so that the gap is exactly $\varepsilon_{\text{step}}$:

$$\eta = \frac{3c+2}{c} (e^{\beta \varepsilon_{\text{step}}} - 1)$$

Next, we prove that a policy is ε -optimal, the policy need at least to chose an optimal arm in a fraction of the bandits. We try then to solve the different bandits with equal probability and the number of steps to do that provide a lower bound.

For a deterministic policy π , let M^π be the number of bandits on which π picks a suboptimal arm and let $\rho = \frac{M^\pi}{n}$. The optimal policy picks the optimal arm for each bandit, hence :

$$V^* = \frac{1}{\beta} \log(1 + cp_+)$$

And:

$$V^\pi = \frac{1}{\beta} \log \left(\frac{(n - M^\pi)(1 + cp_+) + M^\pi(1 + cp_-)}{n} \right) = \frac{1}{\beta} \log \left((1 + cp_+)(1 - \rho + \rho e^{-\beta \varepsilon_{\text{step}}}) \right)$$

Since $e^{-\beta \varepsilon_{\text{step}}} = \frac{1+cp_-}{1+cp_+}$. Hence, the optimality gap verify :

$$V^* - V^\pi = \frac{-1}{\beta} \log \left(1 - (1 - e^{-\beta \varepsilon_{\text{step}}}) \rho \right)$$

If π is ε -optimal then :

$$\rho \leq \frac{1 - e^{-\beta \varepsilon}}{1 - e^{-\beta \varepsilon_{\text{step}}}}$$

This means the policy must at least solve a fraction $\phi = 1 - \frac{1 - e^{-\beta \varepsilon}}{1 - e^{-\beta \varepsilon_{\text{step}}}}$ of the bandits in the MDP to be ε -optimal. We enforce this using a union bound by requiring the per-bandit error to satisfy $\delta_i \leq \delta / (\phi n)$ for those ϕn bandits. This allows to have ε optimality with high probability $1 - \delta$

Fix such a bandit i and the true instance where its unique optimal arm is a_i^* with success probability p_+ . For each suboptimal arm $b \neq a_i^*$, define an alternative MDP $\nu^{(b)}$ that changes only arm b to be the optimal arm. Applying lemma 1 from (Kaufmann et al., 2016) to the event E_b 'the algorithm recommends arm b for bandit i ', since the bandit is solved correctly, we have $\Pr(E_b)$, under instance v , recommending b is a mistake hence $\Pr_v(E_b) \leq \delta_i$ and since b is optimal under $\nu^{(b)}$, recommending b is correct hence gives $\Pr_{\nu^{(b)}}(E_b) \geq 1 - \delta_i$:

$$\mathbb{E}[N_{i,b}] \geq \frac{D_{KL}(1 - \frac{\delta}{\phi n}, \frac{\delta}{\phi n})}{D_{KL}(\text{Ber}(p_-) \parallel \text{Ber}(p_+))}$$

Summing over the $A - 1$ suboptimal arms b ,

$$\mathbb{E}[N_i] = \sum_{a=1}^A \mathbb{E}[N_{i,a}] \geq (A - 1) \frac{D_{KL}(1 - \frac{\delta}{\phi n}, \frac{\delta}{\phi n})}{D_{KL}(\text{Ber}(p_-) \parallel \text{Ber}(p_+))}.$$

Finally, each episode contributes exactly one pull in exactly one bandit, so $T = \sum_{i=1}^n \mathbb{E}[N_i]$. Summing over the ϕn bandits that must be solved and using $D_{KL}(1-x, x) \geq \log(1/(2x))$ (assuming $\delta \leq \frac{\phi n}{2}$ yields :

$$T \geq \phi(S-3)(A-1) \frac{\log\left(\frac{\phi(S-3)}{2\delta}\right)}{D_{KL}(\text{Ber}(p_-) \parallel \text{Ber}(p_+))}$$

And since $D_{KL}(\text{Ber}(p_-) \parallel \text{Ber}(p_+)) \leq \frac{(p_- - p_+)^2}{p_-(1-p_-)} \leq \frac{\eta^2}{2c-1}$ We get :

$$T \geq 4 \frac{(e^{\beta(H-2)} - 1) - 1}{(e^{\beta\varepsilon_{\text{step}}} - 1)^2} \phi(S-3)(A-1) \log\left(\frac{\phi(S-3)}{2\delta}\right)$$

We can pick for example $\varepsilon_{\text{step}} = 2H\varepsilon$ so that for ε small enough:

$$T = \mathcal{O}\left((S-3)(A-1) \frac{2e^{\beta(H-2)} - 3}{\beta^2\varepsilon^2} \log(S/\delta)\right)$$

■

Appendix D. Concentration inequalities

D.1. Sanov's theorem

First we introduce the K divergence concentration inequality derived via Sanov's theorem

Lemma 16 *Let $\Sigma_m = \{q \in \mathbb{R}_{\geq 0}^m : \sum_{i=1}^m q_i = 1\}$ and let $p \in \Sigma_m$. Let $X_1, \dots, X_n$ be i.i.d. with law p , and let the empirical distribution $\tilde{p}_n$ be*

$$\tilde{p}_n(i) = \frac{1}{n} \sum_{k=1}^n \mathbf{1}\{X_k = i\}, \quad i \in [m].$$

Then for any $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$D(\tilde{p}_n \parallel p) \leq \frac{(m-1) \log(n+1) + \log(1/\delta)}{n}.$$

Proof Let $\Sigma_m = \left\{q \in \mathbb{R}_{\geq 0}^m : \sum_{i=1}^m q_i = 1\right\}$ be the $(m-1)$ -simplex and let $p \in \Sigma_m$. Let $X_1, \dots, X_n$ be i.i.d. with law p , and let $\tilde{p}_n$ denote the empirical distribution by

$$\tilde{p}_n(i) := \frac{1}{n} \sum_{k=1}^n \mathbf{1}\{X_k = i\}, \quad i \in [m]$$

Sanov's theorem (Theorem 11.4.1 [Cover and Thomas \(2006\)](#)) states that for any set $E \subseteq \Sigma_m$

$$\Pr(\tilde{p}_n \in E) \leq (n+1)^{(m-1)} \exp\left(-n \inf_{q \in E} D(q \parallel p)\right)$$

Fix $\varepsilon > 0$ and take the set $E = \{q \in \Sigma_m \mid D(q \parallel p) \geq \varepsilon\}$ then $\inf_{q \in E} D(q \parallel p) = \varepsilon$ and by Sanov's theorem:

$$\Pr(D(\tilde{p}_n \parallel p) \geq \varepsilon) \leq (n+1)^m \exp(-n\varepsilon)$$

We turn it to high probability bound by setting the r.h.s to δ which yield:

$$\varepsilon = \frac{m \log(n+1) + \log(1/\delta)}{n}$$

Doing a union bound on all state action pairs and all time steps we get that with probability $1 - \delta$ we have:

$$D(\tilde{p}_n || p) \leq \frac{(m-1) \log(n+1) + \log(SAH/\delta)}{n}$$

■

D.2. Concentration inequality for Bernoulli random variables

We state a deviation inequality for Bernoulli random variables from [Dann et al. \(2017\)](#)(Lemma F.4):

Lemma 17 *Let $\mathcal{F}_i$ for $i = 1 \dots$ be a filtration and $X_1, \dots, X_n$ be a sequence of Bernoulli random variables with $\mathbb{P}(X_i = 1 | \mathcal{F}_{i-1}) = P_i$ with P_i being $\mathcal{F}_{i-1}$ -measurable and X_i being $\mathcal{F}_i$ measurable. It holds that*

$$\mathbb{P} \left(\exists n : \sum_{t=1}^n X_t < \sum_{t=1}^n P_t / 2 - \log\left(\frac{1}{\delta}\right) \right) \leq \delta$$

D.3. Self-normalized Bernstein inequality

We state the self-normalized Bernstein-type inequality by [\(Ménard et al., 2021\)](#).

Lemma 18 *Let $(Y_t)_{t \in \mathbb{N}^*}, (w_t)_{t \in \mathbb{N}^*}$ be two sequences of random variables adapted to a filtration $(\mathcal{F}_t)_{t \in \mathbb{N}}$. We assume that the weights are in the unit interval $w_t \in [0, 1]$ and predictable, i.e. $\mathcal{F}_{t-1}$ measurable. We also assume that the random variables Y_t are bounded $|Y_t| \leq b$ and centered $\mathbb{E}[Y_t | \mathcal{F}_{t-1}] = 0$. Consider the following quantities*

$$S_t \triangleq \sum_{s=1}^t w_s Y_s, \quad V_t \triangleq \sum_{s=1}^t w_s^2 \cdot \mathbb{E}[Y_s^2 | \mathcal{F}_{s-1}], \quad \text{and} \quad W_t \triangleq \sum_{s=1}^t w_s$$

and let $h(x) \triangleq (x+1) \log(x+1) - x$ be the Cramér transform of a Poisson distribution of parameter 1. Then For all $\delta > 0$:

$$\mathbb{P} \left(\exists t \geq 1, (V_t/b^2 + 1)h \left(\frac{b|S_t|}{V_t + b^2} \right) \geq \log(1/\delta) + \log(4e(2t+1)) \right) \leq \delta.$$

The previous inequality can be weakened to obtain a more explicit bound: with probability at least $1 - \delta$, for all $t \geq 1$,

$$|S_t| \leq \sqrt{2V_t \log(4e(2t+1)/\delta)} + 3b \log(4e(2t+1)/\delta).$$

D.4. KL-Bernstein inequality

We state a KL-Bernstein inequality from (Talebi and Maillard, 2018) and (Ménard et al., 2021):

Lemma 19 *Let $p, q \in \Sigma_S$, where Σ_S denotes the probability simplex of dimension $S - 1$. For all $\alpha > 0$, for all functions f defined on S with $0 \leq f(s) \leq b$, for all $s \in S$, if $\text{KL}(p, q) \leq \alpha$ then*

$$|pf - qf| \leq \sqrt{2\text{Var}_q(f)\alpha} + \frac{2}{3}b\alpha$$

where we use the expectation operator defined as $pf \triangleq \mathbb{E}_{s \sim p} f(s)$ and the variance operator defined as $\text{Var}_p(f) \triangleq \mathbb{E}_{s \sim p} (f(s) - \mathbb{E}_{s' \sim p} f(s'))^2 = p(f - pf)^2$.

Appendix E. Technical results

The next lemma introduces the entropic variance under a fixed policy π , defined as the conditional variance of the exponentiated return $e^{\beta G_h}$ around its entropic value $e^{\beta Q_h^\pi}$. It shows that σQ_h^π satisfies a Bellman-style recursion: for each step h and state-action pair (s, a) , $\sigma Q_h^\pi(s, a)$ decomposes into (i) the variance under the next-state transition $p_h(\cdot | s, a)$ of $e^{\beta(r_h(s, a) + V_{h+1}^\pi(s'))}$ and (ii) a continuation term $e^{2\beta r_h(s, a)} p_h \sigma V_{h+1}^\pi(s, a)$, with terminal condition $\sigma_{H+1}^\pi = 0$.

Lemma 20 *For a policy π . Define the entropic variance:*

$$\sigma Q_h^\pi(s, a) = \mathbb{E}^\pi \left[\left(e^{\beta G_h(s, a)} - e^{\beta Q_h^\pi(s, a)} \right)^2 \middle| s_h = s, a_h = a \right]$$

with terminal condition $\sigma_{H+1}^\pi(s, a) = 0$ and:

$$\sigma V_h^\pi(s) = \sigma Q_h^\pi(s, \pi(s))$$

Then $\{\sigma_h^\pi\}_{h=1}^{H+1}$ satisfy the Bellman recursion: For every $h \in [H]$ and state-action pair (s, a) :

$$\sigma Q_h^\pi(s, a) = \text{Var}_{p_h} \left(e^{\beta(r_h(s, a) + V_{h+1}^\pi)} \right)(s, a) + e^{2\beta r_h(s, a)} p_h \sigma V_{h+1}^\pi(s, a)$$

Proof Let h be a step and let (s, a) be a state-action pair. First notice that σQ_h^π is a conditional variance: By the definition of Q_h^π we have:

$$e^{\beta Q_h^\pi(s, a)} = \mathbb{E}^\pi \left[e^{\beta G_h(s, a)} \middle| s_h = s, a_h = a \right]$$

Hence:

$$\begin{aligned} \sigma Q_h^\pi(s, a) &= \mathbb{E}^\pi \left[\left(e^{\beta G_h(s, a)} - \mathbb{E}^\pi \left[e^{\beta G_h(s, a)} \middle| s_h = s, a_h = a \right] \right)^2 \middle| s_h = s, a_h = a \right] \\ &= \text{Var} \left(e^{\beta G_h(s, a)} \middle| s_h = s, a_h = a \right) \end{aligned}$$

Using that the rewards are stationary we have:

$$\text{Var} \left(e^{\beta G_h(s, a)} \middle| s_h = s, a_h = a \right) = e^{2\beta r_h(s, a)} \text{Var} \left(e^{\beta G_{h+1}} \middle| s_h = s, a_h = a \right)$$

Using the law of total variance, we have:

$$\text{Var}\left(e^{\beta G_{h+1}} \middle| s_h = s, a_h = a\right) = \mathbb{E}^\pi \left[\text{Var}\left(e^{\beta G_{h+1}} \middle| S'\right) \middle| s_h = s, a_h = a \right] + \text{Var}\left(\mathbb{E}^\pi \left[e^{\beta G_{h+1}} \middle| S' \right] \middle| s_h = s, a_h = a\right)$$

For the first term we have:

$$\mathbb{E}^\pi \left[\text{Var}(e^{\beta G_{h+1}} \mid S') \mid s_h, a_h \right] = \mathbb{E}^\pi \left[\mathbb{E}^\pi \left[\left(e^{\beta G_{h+1}} - \mathbb{E}^\pi [e^{\beta G_{h+1}} \mid S'] \right)^2 \middle| S' \right] \middle| s_h = s, a_h = a \right]$$

But since by the definition of the value function:

$$e^{\beta V_{h+1}}(s') = \mathbb{E}^\pi [e^{\beta G_{h+1}} \mid s_{h+1} = s']$$

Then:

$$\mathbb{E}^\pi \left[\text{Var}(e^{\beta G_{h+1}} \mid S') \mid s_h, a_h \right] = \mathbb{E}^\pi \left[\mathbb{E}^\pi \left[\left(e^{\beta G_{h+1}} - e^{\beta V_{h+1}}(s') \right)^2 \middle| S' \right] \middle| s_h, a_h \right] = \mathbb{E}_{s' \sim p_h(\cdot \mid s_h, a_h)} \left[\sigma V_{h+1}^\pi(s') \right] = p_h \sigma V_{h+1}^\pi(s, a)$$

For the second term, again using the definition of the value function:

$$\text{Var}\left(\mathbb{E}^\pi \left[e^{\beta G_{h+1}} \middle| S' \right] \middle| s_h = s, a_h = a\right) = \text{Var}_{s' \sim p_h(\cdot \mid s, a)} \left(e^{\beta V_{h+1}}(s') \right) = \text{Var}_{p_h} \left(e^{\beta V_{h+1}}(s') \right)$$

Hence, by replacing both terms we find:

$$\sigma Q_h^\pi(s, a) = \text{Var}_{p_h} \left(e^{\beta(r_h(s, a) + V_{h+1}^\pi)} \right)(s, a) + e^{2\beta r_h(s, a)} p_h \sigma V_{h+1}^\pi(s, a)$$

■

Lemma 21 [A normalized variance bound for an exponential transform] Let $f : \mathcal{X} \rightarrow [0, R]$ and let $\beta \in \mathbb{R}$. Define

$$Y := e^{\beta f(X)}$$

Set

$$m := e^{\min\{0, \beta R\}} = \min\{1, e^{\beta R}\}, \quad M := e^{\max\{0, \beta R\}} = \max\{1, e^{\beta R}\}.$$

Then $Y \in [m, M]$ a.s., $\mu \in [m, M]$, and

$$\frac{\text{Var}(Y)}{\mu^2} \leq \frac{(e^{|\beta|R} - 1)^2}{4e^{|\beta|R}} \leq \frac{e^{|\beta|R} - 1}{4}$$

Proof If $\beta = 0$ then $Y \equiv 1$ and $\text{Var} = 0$, so assume $\beta \neq 0$. Since $f(X) \in [0, R]$, we have $\beta f(X) \in [\min\{0, \beta R\}, \max\{0, \beta R\}]$, hence

$$m \leq Y = e^{\beta f(X)} \leq M \quad \text{a.s.}$$

and therefore $\mu = \mathbb{E}[Y] \in [m, M]$.

By the Bhatia–Davis inequality:

$$\text{Var}(Y) \leq (M - \mu)(\mu - m)$$

Dividing by $\mu^2 > 0$ yields

$$\frac{\text{Var}(Y)}{\mu^2} \leq \frac{(M - \mu)(\mu - m)}{\mu^2}$$

Now consider

$$g(\mu) := \frac{(M - \mu)(\mu - m)}{\mu^2}, \quad \mu \in [m, M]$$

Rewrite

$$g(\mu) = \frac{M + m}{\mu} - \frac{Mm}{\mu^2} - 1, \quad g'(\mu) = -\frac{M + m}{\mu^2} + \frac{2Mm}{\mu^3} = \frac{2Mm - (M + m)\mu}{\mu^3}$$

Thus $g'(\mu) = 0$ iff $\mu^* = \frac{2Mm}{M+m} \in [m, M]$, and g increases on $[m, \mu^*]$ and decreases on $[\mu^*, M]$. Hence the maximum is attained at μ^* . Substituting gives

$$g(\mu^*) = \frac{(M - \frac{2Mm}{M+m})(\frac{2Mm}{M+m} - m)}{(\frac{2Mm}{M+m})^2} = \frac{(M - m)^2}{4Mm}$$

so for all $\mu \in [m, M]$

$$\frac{(M - \mu)(\mu - m)}{\mu^2} \leq \frac{(M - m)^2}{4Mm}$$

Finally, since $M/m = e^{|\beta|R} \geq 1$, we have

$$\frac{(M - m)^2}{4Mm} = \frac{(\frac{M}{m} - 1)^2}{4\frac{M}{m}} = \frac{(e^{|\beta|R} - 1)^2}{4e^{|\beta|R}} \leq \frac{e^{|\beta|R} - 1}{4}$$

because $(e^{|\beta|R} - 1)/(e^{|\beta|R}) \leq 1$ for $e^{|\beta|R} \geq 1$. ■

We state lemma 11 and 12 from (Ménard et al., 2021) that are used for the variance transportation:

Lemma 22 *Let $p, q \in \Sigma_S$ and f is a function defined on $\mathcal{S}$ such that $0 \leq f(s) \leq b$ for all $s \in \mathcal{S}$. If $\text{KL}(p, q) \leq \alpha$ then*

$$\begin{aligned} \text{Var}_q(f) &\leq 2\text{Var}_p(f) + 4b^2\alpha \quad \text{and} \\ \text{Var}_p(f) &\leq 2\text{Var}_q(f) + 4b^2\alpha \end{aligned}$$

Lemma 23 *For $p, q \in \Sigma_S$, for f, g two functions defined on $\mathcal{S}$ such that $0 \leq g(s), f(s) \leq b$ for all $s \in \mathcal{S}$, we have that*

$$\begin{aligned} \text{Var}_p(f) &\leq 2\text{Var}_p(g) + 2bp|f - g| \quad \text{and} \\ \text{Var}_q(f) &\leq \text{Var}_p(f) + 3b^2\|p - q\|_1, \end{aligned}$$

where we denote the absolute operator by $|f|(s) = |f(s)|$ for all $s \in \mathcal{S}$.

We state the pseudo-counts lemma 7 that allows to go from counts to their mean the pseudo-counts and lemma 8 a standard inequality from (Mnardet al., 2021)

Lemma 24 On event $\mathcal{E}^{\text{cnt}}$, for any $\alpha(\cdot, \delta)$ such that $x \mapsto \beta(\delta, x)/x$ is non-increasing for $x \geq 1$, $x \mapsto \beta(x, \delta)$ is non-decreasing $\forall h \in [H], (s, a) \in \mathcal{S} \times \mathcal{A}$.

$$\forall t \in \mathbb{N}^*, \quad \frac{\alpha(n_h^t(s, a), \delta)}{n_h^t(s, a)} \wedge 1 \leq 4 \frac{\alpha(\bar{n}_h^t(s, a), \delta)}{\bar{n}_h^t(s, a) \vee 1}$$

Lemma 25 For $T \in \mathbb{N}^*$ and $(u_t)_{t \in \mathbb{N}^*}$ for a sequence where $u_t \in [0, 1]$ and $U_t \triangleq \sum_{i=1}^t u_i$, we get

$$\sum_{t=0}^T \frac{u_{t+1}}{U_t \vee 1} \leq 4 \log(U_{T+1} + 1).$$

Finally we state lemma 13 from (Ménard et al., 2021)

Lemma 26 Let A, B, C, D, E , and α be positive scalars such that $1 \leq B \leq E$ and $\alpha \geq e$. If $\tau \geq 0$ satisfies

$$\tau \leq C \sqrt{\tau (A \log(\alpha \tau) + B \log(\alpha \tau)^2)} + D (A \log(\alpha \tau) + E \log(\alpha \tau)^2) \quad (3)$$

then

$$\tau \leq C^2(A + B)C_1^2 + \left(D + 2\sqrt{DC}\right)(A + E)C_1^2 + 1$$

where

$$C_1 = \frac{8}{5} \log(11\alpha^2(A + E)(C + D))$$